



Central Coast Regional Water Quality Control Board

May 4, 2026

Sent Via Electronic Mail

Matt Morely
SCRWA Manager
South County Regional Wastewater Authority
Wastewater Treatment and Reclamation Facility
1500 Southside Drive
Gilroy, California, 95020
e-mail: matt.morley@cityofgilroy.org

Dear Matt Morely:

SOUTH COUNTY REGIONAL WASTEWATER TREATMENT AND RECLAMATION FACILITY, 1500 SOUTHSIDE DRIVE, GILROY, SANTA CLARA COUNTY:

- **NOTICE OF APPLICABILITY FOR ENROLLMENT IN ORDER R3-2020-0020, GENERAL WASTE DISCHARGE REQUIREMENTS FOR DISCHARGES FROM DOMESTIC WASTEWATER SYSTEMS WITH FLOWS GREATER THAN 100,000 GALLONS PER DAY; AND**
- **TRANSMITTAL OF MONITORING AND REPORTING PROGRAM R3-2026-0026**

The Central Coast Regional Water Quality Control Board (Central Coast Water Board) reviewed the March 24, 2022 notice of intent to seek authorization for discharges of treated wastewater and available documents associated with South County Regional Wastewater Authority's (SCRWA's) domestic wastewater treatment and reclamation facility (Wastewater System) located at 1500 Southside Drive, Gilroy, Santa Clara County. The Wastewater System, owned by SCRWA, has been regulated pursuant to two permits: (1) Order R3-2017-0028, NPDES CA0049964, waste discharge requirements for discharge to surface water and percolation ponds and (2) Order 98-052, *Master Water Reclamation Requirements for South County Regional Wastewater Authority and Indirect Users* for the treatment and use of recycled water.

On September 25, 2020, the Central Coast Water Board adopted Order R3-2020-0020, *General Waste Discharge Requirements for Discharges from Domestic Wastewater Systems with Flows Greater than 100,000 Gallons per Day* (General Permit). Central Coast Water Board staff has determined that the production of recycled water and the discharge of treated wastewater to the onsite percolation ponds is more appropriately

JANE GRAY, CHAIR | RYAN E. LODGE, EXECUTIVE OFFICER

regulated by the General Permit. The use of recycled water produced by the Wastewater System will be regulated pursuant to the State Water Resources Control Board's (State Water Board's) Order WQ 2016-0068-DDW, *Water Reclamation Requirements for Recycled Water Use* (General Permit 2016-0068-DDW).

Additionally, for emergency situations, SCRWA will remain enrolled in an NPDES permit that regulates the discharge of tertiary treated water to the Pajaro River. Order R3-2026-0014, NPDES CA0049964, was adopted by the Central Coast Water Board on April 16, 2026, becomes effective on July 1, 2026, and supersedes Order R3-2017-0028. SCRWA has not discharged to the Pajaro River since the Wastewater System was first regulated by an NPDES permit in 2004. However, SCRWA has requested to retain the NPDES permit as a contingency discharge location for emergencies, such as storm events, when discharge to the ponds may not be feasible.

This letter serves as a Notice of Applicability for enrollment in the General Permit. This Notice of Applicability is effective on July 1, 2026, concurrent with the effective dates of Order R3-2026-0014 for surface water discharges and enrollment in General Permit 2016-0068-DDW for recycled water.

This Notice of Applicability includes: (1) wastewater system operation summary, specific requirements, and limitations (Attachment 1); (2) figures (Attachment 2); (3) monitoring and reporting program requirements (Attachment 3); and (4) Division of Drinking Water correspondence related to UV disinfection (Attachment 4).

SCRWA must comply with the following:

1. General Permit

- 1.1. SCRWA must comply with all conditions and requirements of the General Permit. As described in the General Permit, ongoing operation, maintenance, monitoring, and reporting are required. A copy of the General Permit can be found electronically at the following link:

https://www.waterboards.ca.gov/centralcoast/board_decisions/adopted_orders/2020/r32020_0020.pdf

2. Monitoring and Reporting Program

- 2.1. SCRWA must comply with the requirements of the monitoring and reporting program enclosed with this Notice of Applicability (Monitoring and Reporting Program R3-2026-0026, Attachment 3). SCRWA must start implementing the monitoring and reporting requirements on **July 1, 2026**. The first quarterly monitoring report is due **November 1, 2026**, and the first annual report is due **April 30, 2027**.

- 2.2. All monitoring reports must be submitted electronically in searchable PDF format, with the Central Coast Water Board's current transmittal sheet used as the cover page. The transmittal sheet must be signed.

https://www.waterboards.ca.gov/centralcoast/water_issues/programs/wastewater_permitting/docs/transmittal_sheet.pdf

- 2.3. The transmittal sheet identifies authorized signatories. To authorize a representative to sign and submit reports on behalf of SCRWA, the "Designation of Duly Authorized Representatives" form must be submitted to the Central Coast Water Board.

https://www.waterboards.ca.gov/centralcoast/water_issues/programs/wastewater_permitting/

- 2.4. All annual reports must be formatted and completed in accordance with the Central Coast Water Board's annual report template found at the following link:

https://www.waterboards.ca.gov/centralcoast/board_decisions/adopted_orders/2020/wdr_annual_report_format.pdf

- 2.5. SCRWA must submit all reports, supporting documents, and laboratory data (using the transmittal sheet as the cover page) to the publicly accessible State Water Board GeoTracker^{1,2} database in accordance with applicable Electronic Submittal of Information (ESI) requirements for Wastewater System-specific global identification number **WDR100034821** at the following link:

<https://geotracker.waterboards.ca.gov/esi/login.asp>

- 2.6. The attached monitoring and reporting program (Attachment 3) describes applicable GeoTracker electronic reporting requirements. The Central Coast Water Board may request submittal of some documents in paper form, particularly drawings or maps that require a large format to be legible, or in alternative electronic formats as needed for data evaluation.

- 2.7. SCRWA is also required to comply with annual volumetric reporting requirements³ established by the State Water Board, due annually to the GeoTracker database by April 30.

¹ Information for first-time GeoTracker users is available at:

https://www.waterboards.ca.gov/ust/electronic_submittal/

² Additional information available at: <http://geotracker.waterboards.ca.gov/>

³ See: [Volumetric Annual Reporting | California State Water Resources Control Board](#)

3. Recycled Water Production

- 3.1. The General Permit authorizes the production of disinfected tertiary recycled water using the existing anthracite tertiary filters and the existing ultraviolet (UV) and chlorine disinfection systems. Use of the membrane bioreactor (MBR) for recycled water production is authorized subject to the conditions set forth in section 3.4.
- 3.2. As specified in the General Permit, any recycled water produced must comply with California Code of Regulations title 22, division 4, chapter 3 sections 60301 through 60355 (Title 22 Water Recycling Criteria); the General Permit; this Notice of Applicability; an approved Title 22 engineering report; and all acceptance conditions established by the Division of Drinking Water. Accordingly, all recycled water produced by SCRWA must be in accordance with its most recent approved Title 22 engineering report and comply all conditions contained in the conditional acceptance letter and spot check bioassay letters most recently issued by the Division of Drinking Water (Attachment 4).
- 3.3. SCRWA updated the Wastewater System to include UV disinfection in 2014 and, with its Notice of Intent, submitted a copy of its September 2016 Title 22 engineering report.⁴ The Division of Drinking Water conditionally accepted the UV system and issued approval letters, including follow-up spot-check bioassay requirements, dated February 6, 2019, and December 8, 2020. SCRWA must comply with all conditions contained in Division of Drinking Water spot-check bioassay letters (Attachment 4). A subset of these requirements is included in this Notice of Applicability as Attachment A. SCRWA was not able to locate a copy of a previous Title 22 engineering report or Division of Drinking Water conditional acceptance letters. Only acceptance of the 2014 UV system is documented.
- 3.4. SCRWA is currently constructing a membrane bioreactor (MBR) for recycled water production. However, the MBR process has not been accepted by the Division of Drinking Water. This treatment process must be accepted by the Division of Drinking Water prior to its use for recycled water production. Upon receiving acceptance from the Division of Drinking Water, SCRWA may use the MBR to produce recycled water in compliance with the Division of Drinking Water's conditional acceptance letter and with notification to the Central Coast Water Board.

⁴ A copy of the [Title 22 Engineering Report for UV Disinfection](https://geotracker.waterboards.ca.gov/profile_report.asp?global_id=WDR100034821) is available on GeoTracker under the Site Maps/Documents tab:
https://geotracker.waterboards.ca.gov/profile_report.asp?global_id=WDR100034821

- 3.5. On March 13, 2026, SCRWA submitted an updated draft Title 22 engineering report. Central Coast Water Board staff requested the new Title 22 engineering report to replace missing historical documents and to document the new MBR process. After SCRWA receives comments on the draft engineering report, SCRWA must submit a final Title 22 engineering report documenting the design and operation of all components of the recycled water system, including the full tertiary treatment process and disinfection processes, as described in the attached monitoring and reporting program.

4. Fees

- 4.1. SCRWA paid an annual fee of \$68,183 on January 2, 2026, for the existing NPDES permit and \$4,300 for Master Water Reclamation Requirements Order 98-052. This payment covers SCRWA's enrollment in the General Permit for the 2025-2026 fiscal year.
- 4.2. SCRWA must pay an annual fee to maintain coverage in the General Permit. Annual fees are determined by the State Water Board's fee program and cover the state fiscal year of July 1 through June 30. Fees are charged (and updated) annually and are based on the facility's threat and complexity rating and contain a surcharge for water recycled as described in the fee schedule. SCRWA's current annual fee is \$46,915 plus \$2,800 for production of recycled water. SCRWA's Wastewater System is currently assigned a threat and complexity rating of 2A, and the Wastewater System is a producer of recycled water. A copy of the current state fee schedule is available electronically at the following link:

https://www.waterboards.ca.gov/resources/fees/water_quality/#wdr

5. Notification

- 5.1. The Central Coast Water Board will be notified of SCRWA's enrollment in the General Permit at a regularly scheduled public meeting on June 17-18, 2026. Details about that meeting are available on our website at:

https://www.waterboards.ca.gov/centralcoast/board_info/agendas/

6. Future Discharge Modifications

- 6.1. Pursuant to California Water Code section 13260, SCRWA must inform the Central Coast Water Board at least 120 days prior to modifying its discharge. If there are any significant changes in treatment or disposal methodologies or the volume or character of the treated wastewater, SCRWA must notify the Central Coast Water Board immediately of such changes.

- 6.2. Attachment 1 contains the Central Coast Water Board's understanding of the Wastewater System. Please review Attachment 1 to determine if the description is representative of the current system. Pursuant to California Water Code section 13267, SCRWA must notify the Central Coast Water Board if this information is out of date or incorrect. SCRWA is required to provide the Central Coast Water Board with updated information within **30 days** of the effective date of this Notice of Applicability.

7. Responsible Party

- 7.1. SCRWA is responsible for the management and disposal of the domestic wastewater in compliance with the conditions of the General Permit. Any noncompliance with this General Permit constitutes a violation of the California Water Code and subjects SCRWA to enforcement action and termination of enrollment under the General Permit.

8. Change in Ownership

- 8.1. In the event of any change in control or ownership of the property, SCRWA must notify the succeeding owner or operator of the existence of the General Permit by letter, a copy of which shall be immediately forwarded to the Central Coast Water Board so that the new owner or operator can be enrolled in the General Permit and SCRWA's enrollment in the General Permit can be terminated.

9. Certified Operator

- 9.1. SCRWA is required to comply with the operator certification requirements⁵ administered by the State Water Board's Office of Operator Certification, which currently requires a Grade V certified operator to maintain and operate the Wastewater System. SCRWA must:
- 9.1.1. Keep a current Wastewater Treatment Plant Classification Form on file with the Office of Operator Certification and update the form when modifications to the Wastewater System treatment processes occur.
 - 9.1.2. Provide the Office of Operator Certification with a Chief Plant Operator Acknowledgement Form and update this information within 30 days of a change of chief plant operator.
 - 9.1.3. If a contract operator is used, the contract operator must register with the Office of Operator Certification and obtain a contract operator certificate for the Wastewater System.
 - 9.1.4. Forms are available at the following link:

⁵ Title 23 division 3, chapter 26, Wastewater Treatment Plant Classification, Operator Certification and Contract Operator Registration.
https://www.waterboards.ca.gov/water_issues/programs/operator_certification/docs/ocr_clean.pdf

https://www.waterboards.ca.gov/water_issues/programs/operator_certification/form.html

10. Disposal Area

10.1. Discharge to areas other than the designated percolation ponds shown in Figure 2 of Attachment 2 is prohibited, except to the Pajaro River in compliance with the NPDES permit and recycled use areas in compliance with Title 22 Water Recycling Criteria. If recycled water is not adequately treated per Title 22 Water Recycling Criteria, off-specification water must be disposed of as described in the Title 22 engineering report contingency plan.

11. Compliance Schedule

11.1. Reported water quality data indicate that concentrations of total dissolved solids, chloride, sodium, sulfate, and boron in effluent exceed the *Water Quality Control Plan for the Central Coastal Basin* (Basin Plan) water quality objectives for the Pajaro Valley Basin Llagas Sub-Area (Designated Groundwater Basin). However, SCRWA can comply with the General Permit by meeting the effluent limitations of its previous permit (Order R3-2017-0028) for up to two years, as specified in section 3.3 of Attachment 1. As detailed in the attached monitoring and reporting program, SCRWA must prepare and submit a time schedule compliance plan for Central Coast Water Board review and approval by July 1, 2028.

If you have any questions, please contact **Lauren Sipich at (805) 549-3504 or by email at Lauren.Sipich@waterboards.ca.gov** or Leah Lemoine at (805) 549-3159 or by email at Leah.Lemoine@waterboards.ca.gov.

Sincerely,

Leah

Lemoine

for Ryan E. Lodge
Executive Officer

The logo for the Water Boards, featuring three stylized blue waves of increasing height from left to right.

Digitally signed by Leah
Lemoine

Date: 2026.05.01

13:12:04 -07'00'

Attachments

Attachment 1. Wastewater System Operation Summary, Specific Requirements, and Limitations

Attachment 2. Figures

Attachment 3. Monitoring and Reporting Program R3-2026-0026

Attachment 4. Division of Drinking Water Correspondence related to UV Disinfection

cc:

Bret Swain, South County Regional Wastewater Authority, bret.swain@cityofgilroy.org
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WDR Program, GeoTracker, RB3-WDR@Waterboards.ca.gov

ECM/CIWQS = CW-256038

GeoTracker = GT- WDR100034821

Rev 6/25/25

ECM Subject Name = SCRWA NOA for R3-2020-0020

[https://cawaterboards.sharepoint.com/rb3/wdr/documents/wdr facilities/north counties/scrwa/r3_2020_0020_draft_noa_mrp_scrwa_for discharger_all_comments_2026.04.06.docx](https://cawaterboards.sharepoint.com/rb3/wdr/documents/wdr%20facilities/north%20counties/scrwa/r3_2020_0020_draft_noa_mrp_scrwa_for%20discharger_all_comments_2026.04.06.docx)

**ATTACHMENT 1
WASTEWATER SYSTEM OPERATION SUMMARY, SPECIFIC REQUIREMENTS,
AND LIMITATIONS**

1. Ownership and Wastewater System Information

1.1. Ownership and wastewater system information submitted by SCRWA to the Central Coast Water Board is shown in Table 1.

Table 1. Ownership and Wastewater System Information

Name	South County Regional Wastewater Treatment and Water Reclamation Facility
Physical Address	1500 Southside Drive, Gilroy, CA 95020
Latitude	36.98276
Longitude	-121.53284
Mailing Address	1500 Southside Drive, Gilroy, CA 95020
Owner	South County Regional Wastewater Authority
Operator	John Edgar (Jacobs) phone: (408) 318-4660 e-mail: John.Edgar@jacobs.com
Legally Responsible Official	Matt Morely phone: (408) 848-0480 e-mail: matt.morley@cityofgilroy.org
Billing Contact	City of Gilroy 1500 Southside Drive Gilroy, California, 95020 phone: (408) 848-0480 email: matt.morley@cityofgilroy.org
Service Area	Sewer Connections: 28,309 Population Served: 105,003 Domestic: 85% Commercial/Industrial: 15%
Permitted Design Flow, Average Dry Weather Flow (million gallons per day; MGD)	8.5; average dry-weather flow treatment capacity for disposal to percolation ponds 11; average dry-weather flow treatment capacity for disposal to percolation ponds after the MBR is brought online
Permitted Recycled Water Flow, Maximum Daily MGD	9.0
Baseline Flow (2025 average annual influent flow) in MGD	6.59
Threat to Water Quality	2
Complexity	A
For Internal Use	

Fee Code	58
Primary Place Type	Wastewater Treatment Facility
Facility Type	Municipal/Domestic
Facility Waste Type	Domestic Wastewater
Regulatory measure Type	Enrollee - WDR
Reclamation Included	Yes, Producer non-potable
EPA Approved Pretreatment Program	No

2. Wastewater System Operation Summary

- 2.1. The Wastewater System provides municipal wastewater treatment and disposal for the cities of Gilroy and Morgan Hill. Beginning in the mid-1970's through April 1995, the Wastewater System used treatment via aerated ponds. During this time, the Wastewater System consisted of two subsystems: one for treatment and disposal of municipal wastewater from Gilroy and Morgan Hill that was managed by SCRWA and the second for treatment and disposal of food processor wastewater that was managed by Gilroy. In 1997, treatment of the food process wastewater in the aerated ponds was discontinued. In 1999, SCRWA purchased the ponds formerly used for food processor wastewater and converted these ponds to municipal use.
- 2.2. Currently, the Wastewater System consists of pre-anoxic basins, oxidation ditches, post-anoxic basins, a reaeration basin, and secondary clarifiers along with a tertiary treatment plant with anthracite media filters and disinfection. These processes are described in detail in Table 2. A membrane bioreactor (MBR) with additional headworks is being constructed for parallel operation with the existing oxidation ditches and clarifiers and will be permitted to operate after completion and startup testing.
- 2.3. In its notice of intent, submitted on March 22, 2022, SCRWA states that the Wastewater System can treat 8.5 gallons per day (MGD) as average dry weather flow and 10.8 MGD average wet-weather flow to secondary standards. SCRWA states that the Wastewater System can treat 9.0 million gallons per day (MGD) to a Title 22 disinfected tertiary recycled water level. However, the Wastewater System is expanding to include a membrane bioreactor that will increase the Wastewater System capacity to secondary standards by 2.5 MGD to 11 MGD average dry weather flow and 13.3 MGD average wet-weather flow for disposal to the percolation ponds. The Central Coast Water Board is permitting the operation and disposal of treated effluent as described below. SCRWA must not change operation, treatment technology, or disposal methods without Central Coast Water Board notification and review and Division of Drinking Water review and approval for recycled water production components.
- 2.4. The Wastewater System is a Class V wastewater system. SCRWA must staff a Grade V chief plant operator. As of the writing of this Notice of Applicability, the current chief plant operator is John Edgar, Grade V, license number 42059.

2.5. The Wastewater System consists of the treatment technologies and treatment capabilities shown in Table 2. The Wastewater System location, facility map, and process flow diagram are depicted in figures 1 through 4 of Attachment 2.

Table 2. Treatment Technology

Headworks/Preliminary Treatment	Preliminary treatment consists of six raw influent pumps at the influent pump station, two automatic bar racks, two washer compactors, and an odor scrubber. Headworks consists of two alternating bar screens with a third manual screen receiving influent from the influent pump station and the septage receiving station. Screened solids are conveyed through a washer/compactor for offsite disposal. Screened effluent flows into an aerated grit removal chamber system with air-lift grit pumps, grit classifier, vortex grit separator, and odor scrubber. The headworks was recently upgraded to divert flow to the membrane bioreactor and provide fine screening through two rotary drum screens to protect the downstream membrane equipment. The main influent pump station at the Southside Drive headworks has a maximum daily flow capacity of 45 MGD, of which up to 6 MGD can be diverted to the MBR headworks.
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Secondary Treatment (Two processes available)	Secondary anoxic/oxic treatment consists of two treatment trains running in parallel. Each treatment train has six pre-anoxic basins, an oxidation ditch, and six anoxic recycle pumps that return oxidized wastewater back to the pre-anoxic basins. Treated oxidized effluent from each treatment train is combined into a post-anoxic basin followed by a reaeration basin. Treated wastewater is split and clarified through one of three clarifiers that run in parallel. Six pumps transfer return activated sludge (RAS) back to the pre-anoxic basins or to waste. Clarified wastewater is sent to a filter pump station for either disposal or further treatment. This process can treat 8.5 MGD average dry weather flow. A new membrane bioreactor (MBR) operates in parallel with the existing oxidation ditches, clarifier, and anthracite filter system. However, the MBR has not been approved by Division of Drinking water as of the date of this Notice of Applicability; therefore, the MBR may not be used for recycled water production until the Division of Drinking Water provides acceptance. Wastewater treated by the MBR may be disposed of in the percolation ponds. The MBR consists of two membrane bioreactor process trains that include anoxic/aeration zones, separate submerged membrane tanks, two process air blowers, and two membrane backwash pumps. Operation of the MBR will increase the secondary treatment capacity to 11 MGD average dry weather flow.
Tertiary Treatment	Tertiary treatment consists of four anthracite gravity filters, two backwash pumps, four filter feed pumps, three utility water pumps, and five reclamation pumps. An MBR operates in parallel and conveys treated water to ultraviolet (UV) disinfection. The current tertiary treatment capacity is 9 MGD average dry weather flow. The MBR must not be used for recycled water production until the MBR process has been reviewed and approved by Division of Drinking Water. The average dry weather flow of the MBR is 2.5 MGD.

Disinfection (Two processes available)	Disinfection for tertiary treatment is attained through a chlorine contact basin (8 MGD average dry weather flow) or a two-channel low-pressure high-intensity ultraviolet (UV) reactor system (9 MGD average dry weather flow). Both systems are available; however, UV disinfection will be utilized as the primary form of disinfection for water quality purposes. The UV system effluent can be pumped to the chlorine contact basin. The chlorine contact basin may act as an effluent basin when UV is operating as the disinfection process.
Wastewater Disposal Method	Wastewater can be disposed of via 37 percolation ponds with a combined area of 395 acres and an estimated annual wet season average annual disposal capacity of 11 MGD. Each percolation pond covers between 5 to 25 acres. The total pond storage capacity is approximately 320 million gallons. Effluent is applied 6 to 12 inches deep so that water infiltrates within 12 days of application, and then the disposal pond rests for at least two days before re-application. Recycled water that does not meet Title 22 Recycled Water Criteria may be disposed of in the percolation ponds as a contingency.
Biosolids/Sludge Production and Handling	Biosolids and sludge are processed/dewatered via belt filter press, which produces approximately 2,500 dry tons annually. Biosolids are disposed at: Synagro Central LLC 13757 South Harmon Road Dos Palos, CA 93620 – Tel (209) 725-2828 or Liberty Composting, Inc 12421 Holloway Road PO BOX 309 Lost Hills, California 93249-0005
Non-Potable Recycled Water Production	Permitted to produce up to 9.0 MGD of tertiary disinfected recycled water. The recycled water storage reservoir is lined and covered and has a capacity of 3 million gallons.

2.6. A summary of historic and recent average effluent concentrations is included in Table 3. The Central Coast Water Board used reported effluent water quality data to establish the effluent and groundwater limitations in the Wastewater System-Specific Requirements and Limitations section (Section 3).

Table 3. Effluent Water Quality for Select Constituents

Constituents	Units	Annual Average for 2023 ^[1]	Annual Average for 2024 ^[1]	Annual Average for 2025 ^[1]
Biochemical Oxygen Demand, 5-Day	mg/L ^[2]	3.76	4.97	4.13
Total Suspended Solids	mg/L	2.46	2.85	2.35
Total Dissolved Solids	mg/L	607.50	610	680
Chloride	mg/L	192.50	190	200
Sodium	mg/L	130	122.50	130
Sulfate	mg/L	62	57.25	60.67
Boron	mg/L	0.36	0.25	0.21
TKN	mg/L	2.63	1.41	1
Nitrite	mg/L	1	1	1
Total Nitrogen	mg/L	6.59	4.17	4.28
pH	pH units	7.42	7.41	7.42
Nitrate as Nitrogen	mg/L	2.97	1.77	2.28

[1] Based on water quality data reported by SCRWA by email dated March 13, 2026.

[2] mg/L denotes milligrams per liter

3. Wastewater System Specific Requirements and Limitations

3.1. SCRWA must operate the Wastewater System in accordance with the General Permit and this Notice of Applicability. SCRWA must comply with the wastewater system-specific limitations specified in Table 4 through Table 7.

Table 4. Flow Limitations

Flow	Units	Limit	Point of Compliance ^[1]
Average Wet-Weather Flow Effluent Disposal to Ponds	MGD ^[2]	13.3 ^{[3] [4]}	EFF-001
Average Dry-Weather Flow Effluent Disposal to Ponds	MGD	11 ^{[4] [5]}	EFF-001
Maximum Daily Recycled Water Production ^[6]	MGD	9.0	EFF-003

[1] Point of compliance shown on Figure 4.

[2] MGD denotes million gallons per day.

[3] Value based on average daily wet-weather flow capacity after the MBR has been brought online.

[4] This is the permitted capacity once the wastewater system upgrade is complete and Discharger has provided notification to the Central Coast Water Board (average dry weather secondary treatment capacity). Until the MBR upgrade is complete, the average wet-weather flow influent

and the average wet-weather flow effluent limit to the ponds is 10.8 MGD and the average dry-weather flow average is 8.5 MGD.

- [5] Value based on average daily dry weather flow capacity.
[6] The MBR must not be used for recycled water production until Division of Drinking water accepts the MBR and notification is made to the Central Coast Water Board.

3.2. Effluent Limitations for Land Discharge:

3.2.1. In accordance with the General Permit, SCRWA must comply with the effluent limitations and operational parameters specified in Table 5, as well as the recycled water producer requirements specified in Table 7. All wastewater must be treated to disinfected tertiary recycled water treatment standards.

Table 5. Effluent Limitations: Enhanced Activated Sludge and Tertiary Treatment

Constituent	Units	30-Day Average	7-Day Average	Sample Maximum	Point of Compliance
Biochemical Oxygen Demand, 5-Day	mg/L ^[1]	30	45	Not Applicable	EFF-001
Total Suspended Solids	mg/L	30	45	Not Applicable	EFF-001
Settleable Solids	mL/L ^[2]	0.1	0.3	0.5	EFF-001
Biochemical Oxygen Demand, 5-Day	mg/L	10	Not Applicable	20	EFF-003
Total Suspended Solids	mg/L	10	Not Applicable	20	EFF-003
Total Nitrogen	mg/L	10	Not Applicable	Not Applicable	EFF-001 and EFF-003
pH	Not Applicable	Not Applicable	Not Applicable	Between 6.5 and 8.4	EFF-001 and EFF-003

[3] mg/L denotes milligrams per liter

[4] mL/L denotes milliliters per liter

3.3. Interim Effluent Limitations

3.3.1. Discharger-reported water quality data (see Table 3) indicate that concentrations of total dissolved solids, chloride, sodium, sulfate, and boron in effluent exceed the *Water Quality Control Plan for the Central Coastal Basin* (Basin Plan) water quality objectives for the Pajaro Valley Basin Llagas Sub-Area (Designated Groundwater Basin). However, SCRWA can maintain historical effluent limitations as interim effluent limitations, under the General Permit and this Notice of Applicability, by meeting the effluent limitations of its previous permit (Order R3-2017-0028) for up to two years, as shown in Table 6. **The groundwater limitations**

shown in Table 8 will apply in 24 months from the effective date of this Notice of Applicability.

Table 6. Interim Effluent Limitations for EFF-001 ^[1] – Effective until June 30, 2028

Constituents	Units	Average Monthly
Total Dissolved Solids	mg/L ^[2]	900
Chloride	mg/L	200
Sodium	mg/L	175
Sulfate	mg/L	150
Boron	mg/L	1.0

[1] Effluent limitations established by Order R3-2017-0028, which additionally permitted discharge to land.

[2] denotes milligrams per liter.

3.4. Recycled Water Limitations

- 3.4.1. SCRWA produces disinfected tertiary recycled water for off-site use and must therefore comply with Title 22 Water Recycling Criteria, including the limitations and operating requirements in Table 7. SCRWA must comply with all recycled water requirements on the days SCRWA is producing recycled water.
- 3.4.2. SCRWA must comply with the facility-specific UV disinfection requirements provided in the Division of Drinking Water's spot check bioassay letter dated February 1, 2018, and the updates to those conditions included in its subsequent letters dated February 6, 2019, and December 8, 2020. The Division of Drinking Water approved the alarms associated with the UV system in its email dated June 24, 2021. SCRWA must also comply with any future updates, revisions, or additional requirements issued by Division of Drinking Water related to UV disinfection. A subset of the UV disinfection limitations and operational parameters are included in Table 7.
- 3.4.3. SCRWA may not use the MBR for recycled water production until the process has been reviewed and approved by the Division of Drinking Water.
- 3.4.4. Per articles 8 and 10 of the Title 22 Water Recycling Criteria, SCRWA must always maintain the reliability features and contingency measures for the Wastewater System processes and ensure non-compliant recycled water is not being delivered for recycled water use.
- 3.4.5. Per Title 22 section 60329(d), any discharge of untreated or partially treated wastewater to the use area, and the cessation of same, must be reported immediately by telephone to the Central Coast Water Board, Division of Drinking Water, and the local health officer.

Table 7. Limitations for Non-Potable Recycled Water

Constituent or Operational Parameter	Units	Sampling Frequency	Regulatory Limitations	Point of Compliance
Total Coliform	MPN / 100 mL ^[1]	Grab, Daily	Effluent must not exceed: a. Seven-day median concentration of 2.2 MPN/100mL b. 23 MPN/100mL in more than one sample taken over a 30-day range c. 240 MPN/100mL at any time	EFF-003
Turbidity	NTU ^[2]	Continuous	Effluent after filtration must not exceed: a. An average of 2 NTU within a 24-hour period b. 5 NTU more than 5 percent of the time within a 24-hour period; and c. 10 NTU at any time	FIL-001
CT Value ^{[3] [4]}	mg/L*min ^[5]	Calculated, based on total chlorine residual	The chlorine disinfection process following filtration must provide a CT value of not less than 450 milligram-minutes per liter at all times with a modal contact time of at least 90 minutes, based on peak dry weather design flow.	CCB-001
Ultraviolet (UV) Dose ^{[6] [7]}	mJ/cm ²	Continuous	Each UV channel must be operated independently to deliver a minimum UV dose of 112 mJ/cm ² at all times ^[7]	UVS-001
UV Sensor Intensities	mW/cm ² ^[8]	Continuous	2.0 to 5.3 ^[9]	UVS-001

Constituent or Operational Parameter	Units	Sampling Frequency	Regulatory Limitations	Point of Compliance
UV Disinfection Flow ^[7]	MGD	Continuous	Total UV system flow is limited to 9.0 MGD 4.5 MGD per UV flow channel ^[7]	UVS-001 UVS-001A and UVS-001B

[1] MPN/100 mL denotes most probable number per 100 milliliters

[2] NTU denotes nephelometric turbidity units

[3] CT Value is the product of total chlorine residual concentration (mg/L) and modal contact time (minutes) measured at the same point, per CCR title 22 section 60301.230

[4] monitoring required when operating chlorine disinfection process

[5] mg/L*min denotes milligram minutes per liter and is the product of total chlorine residual and modal contact time

[6] The UV dose must be calculated using the facility-specific equation approved by the Division of Drinking Water as described in section 3.5.

[7] Limitation established in Division of Drinking Water's letter dated February 6, 2019.

[8] mW/cm² denotes milliwatts per square centimeter

[9] Limitation established in Division of Drinking Water's letter dated December 8, 2020.

3.5. SCRWA must comply with the facility-specific UV disinfection requirements recommended in Division of Drinking Water's letters dated February 1, 2018, and February 6, 2019, and all subsequent Division of Drinking Water conditional approvals and requirements for the UV spot-check bioassays. Currently, the equations below must be used for each UV reactor as part of the automatic UV disinfection control system for calculating UV dose⁶ and based on the February 1, 2018 and February 6, 2019 Division of Drinking water letters:

$$S_o = e^{(-0.63)} \times e^{0.0452 \times UVT} \times Power^{0.9124}$$

$$RED_{calc} = CR \times 10^{1.863} \times UVA^{-1.615} \times [S/S_o]^{0.927} \times Q^{-0.839}$$

Where:

UVT = UV transmittance at 254 nm, expressed as a whole number, e.g., 60 to 72

S = Measured UV sensor value (mW/cm²)

S_o = Calculated intensity from new lamp at full power (at same UVT) with clean sleeves, typically expressed as a function of UVT (mW/cm²)

RED_{calc} = UV dose calculated per bank (mJ/cm²)

CR = Confidence factor = 0.938

UVA= UV absorbance at 254 nm (cm⁻¹), e.g., between 0.222 and 0.142

Q = Flow rate, calculated as gallons per minute divided by the number of lamps in one bank (gpm/lamp)

Power = Measured ballast power setting (kW)

⁶ Values from the "Bioassay Testing Results for the SCRWA WEDECO TAK55 UV System," final report dated July 2017.

- 3.6. The UV system must be operated in accordance with an approved operations plan (current plan dated January 2019) that specifies clearly the operational limits and response required for critical alarms. Any updates or modifications to the operations plan must be submitted to, and approved by, Division of Drinking Water prior to implementation.

4. Groundwater Limitations

- 4.1. Reported water quality data (see Table 3) indicate that concentrations of total dissolved solids, chloride, sodium, sulfate, and boron in effluent exceed the Basin Plan water quality objectives for the Pajaro Valley Basin Llagas Sub-Area. Therefore, SCRWA's compliance point for these constituents is underlying groundwater, as specified in General Permit section V.A.2.ii.
- 4.2. As required by General Permit, the discharge shall not cause underlying groundwater to exceed water quality objectives set forth in the Basin Plan. Specifically, the Wastewater System must comply with water quality objectives for groundwater found in section 3.3.4 of the Basin Plan, including those for organic chemicals, inorganic chemicals, and radionuclides, which are established at their respective drinking water maximum contaminant levels (MCLs) as defined in California Code of Regulations title 22, division 4, chapter 15.
- 4.3. Specific objectives for salts and nitrogen for the Pajaro Valley Basin Llagas Sub-Area, as listed in Table 8.

Table 8. Groundwater Limitations for Designated Groundwater Basin – Pajaro Valley Basin Llagas Sub-Area, Effective July 1, 2028

Constituents	Units	25-Month Rolling Median ^[1]
Total Dissolved Solids	mg/L ^[2]	300
Chloride	mg/L	20
Sodium	mg/L	20
Sulfate	mg/L	50
Boron	mg/L	0.2
Total Nitrogen ^[3]	mg/L	5

[1] Rolling median is the median of all data collected over the most recent 25 months. When the 25-month rolling median uses historical data that pre-dates this Notice of Applicability, the calculated 25-month rolling median may be qualified.

[2] denotes milligrams per liter.

[3] Compliance for total nitrogen will be evaluated and monitored at EFF-001 and EFF-003.

5. Pond Requirements

- 5.1. SCRWA's wastewater system includes storage and disposal ponds. Therefore, SCRWA must comply with General Permit section IV.B specifications for pond systems, including storage and disposal.

6. Setback Requirements

- 6.1. General Permit section IV.A.10, Table 2, includes setback requirements for the wastewater treatment system, land application, and wastewater impoundments (including ponds). SCRWA disposes of wastewater to percolation ponds; therefore, SCRWA must comply with the wastewater system, land application, and wastewater storage and/or treatment ponds setback requirements outlined in Table 2 of the General Permit.
- 6.2. The disposal ponds and their proximity to Llagas Creek are depicted on Figure 5. Portions of ponds E1 through E7 and S1 through S9, are adjacent to, or potentially within the 150-foot required setback, of Llagas Creek. Pursuant to General Permit section IV.A.10, the Executive Officer may approve a reduced setback based on site-specific conditions. SCRWA previously submitted a memorandum dated May 21, 2009, which characterizes Llagas Creek as a losing stream with limited, if any, hydraulic connectivity to the disposal area. Based on this information, a reduced setback is permitted provided that continued surface water monitoring is conducted to verify that the disposal of treated effluent into ponds will have no impact on surface water or beneficial uses.

ATTACHMENT 2
FIGURES

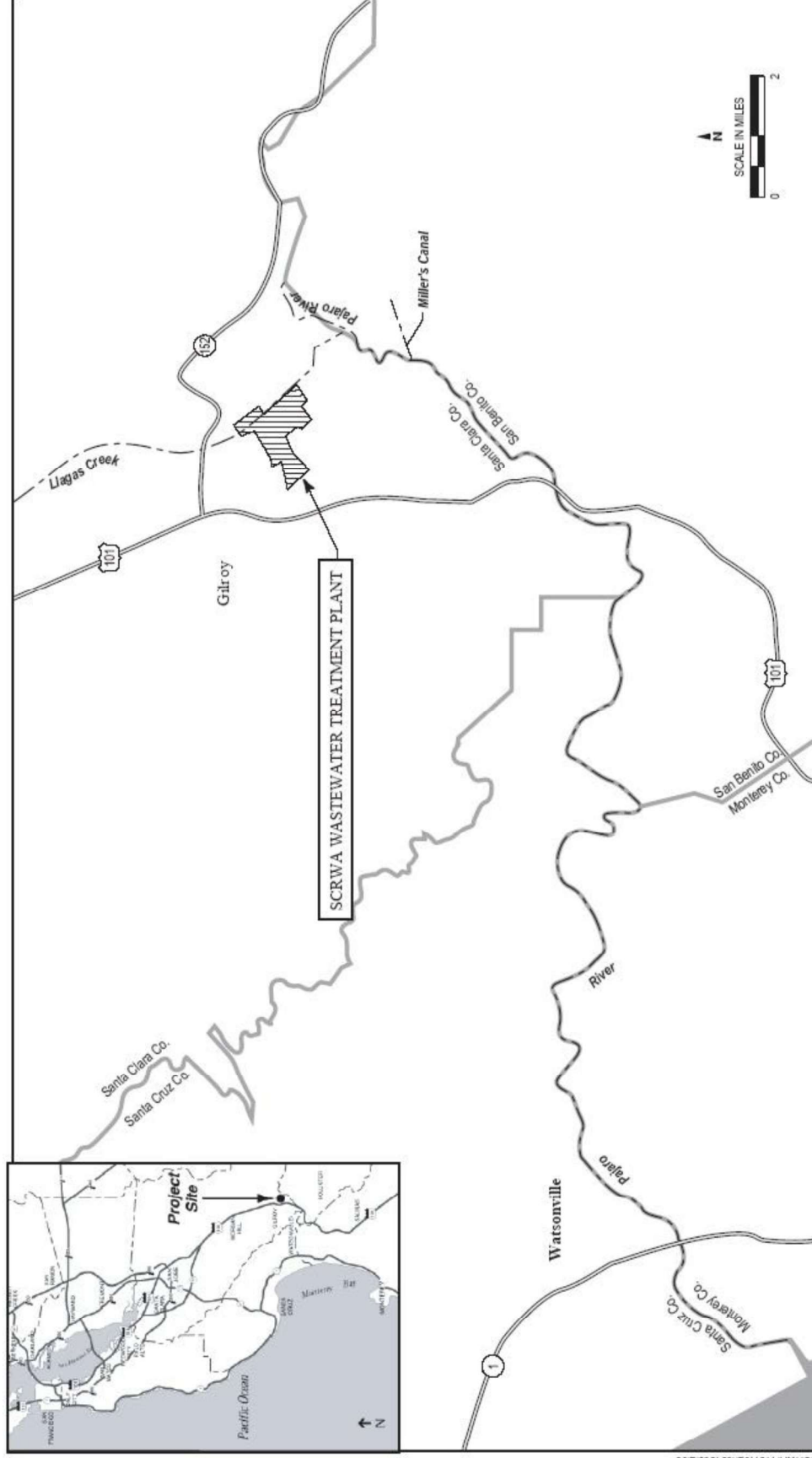


FIGURE 1. FACILITY LOCATION MAP

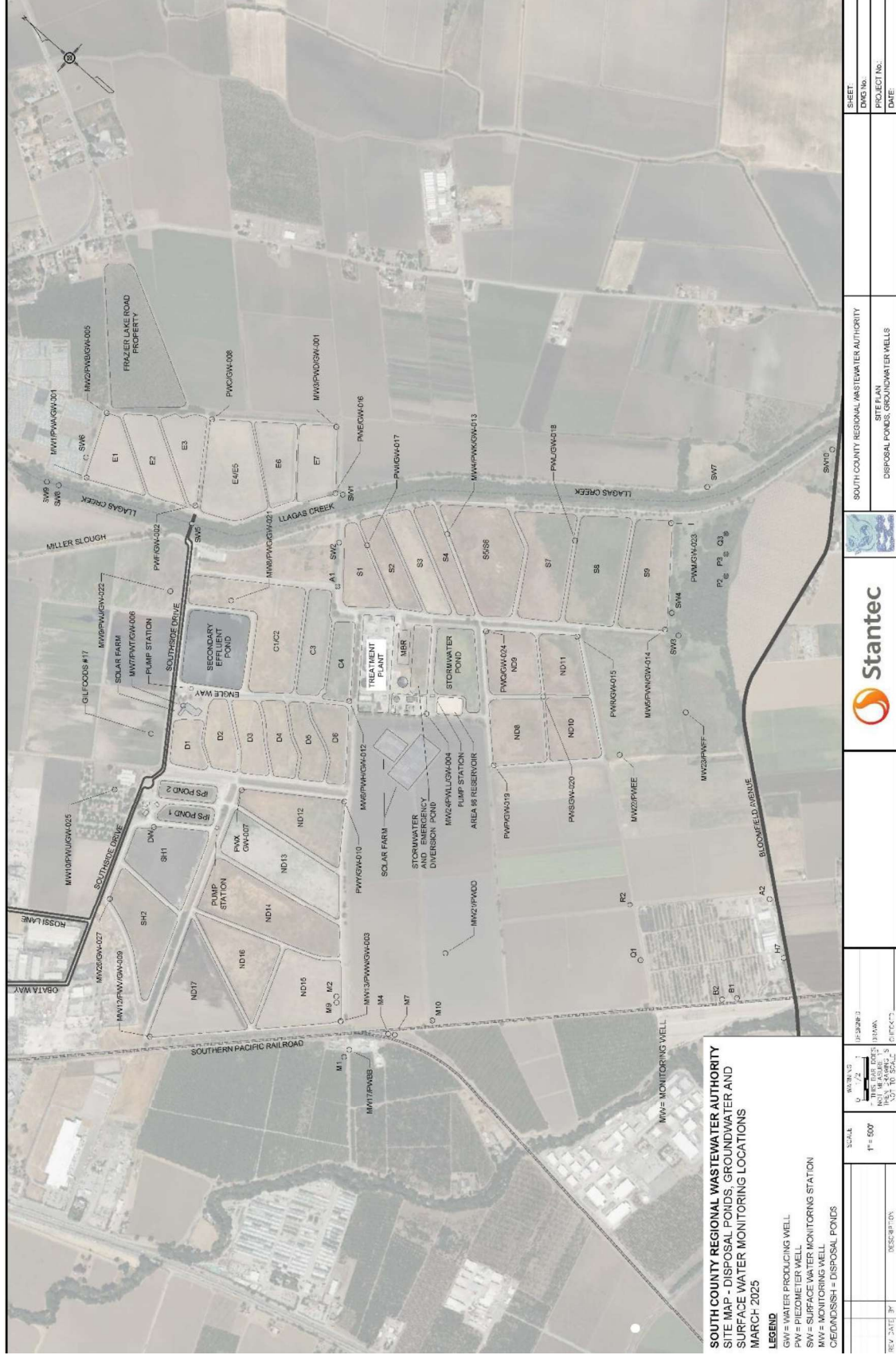
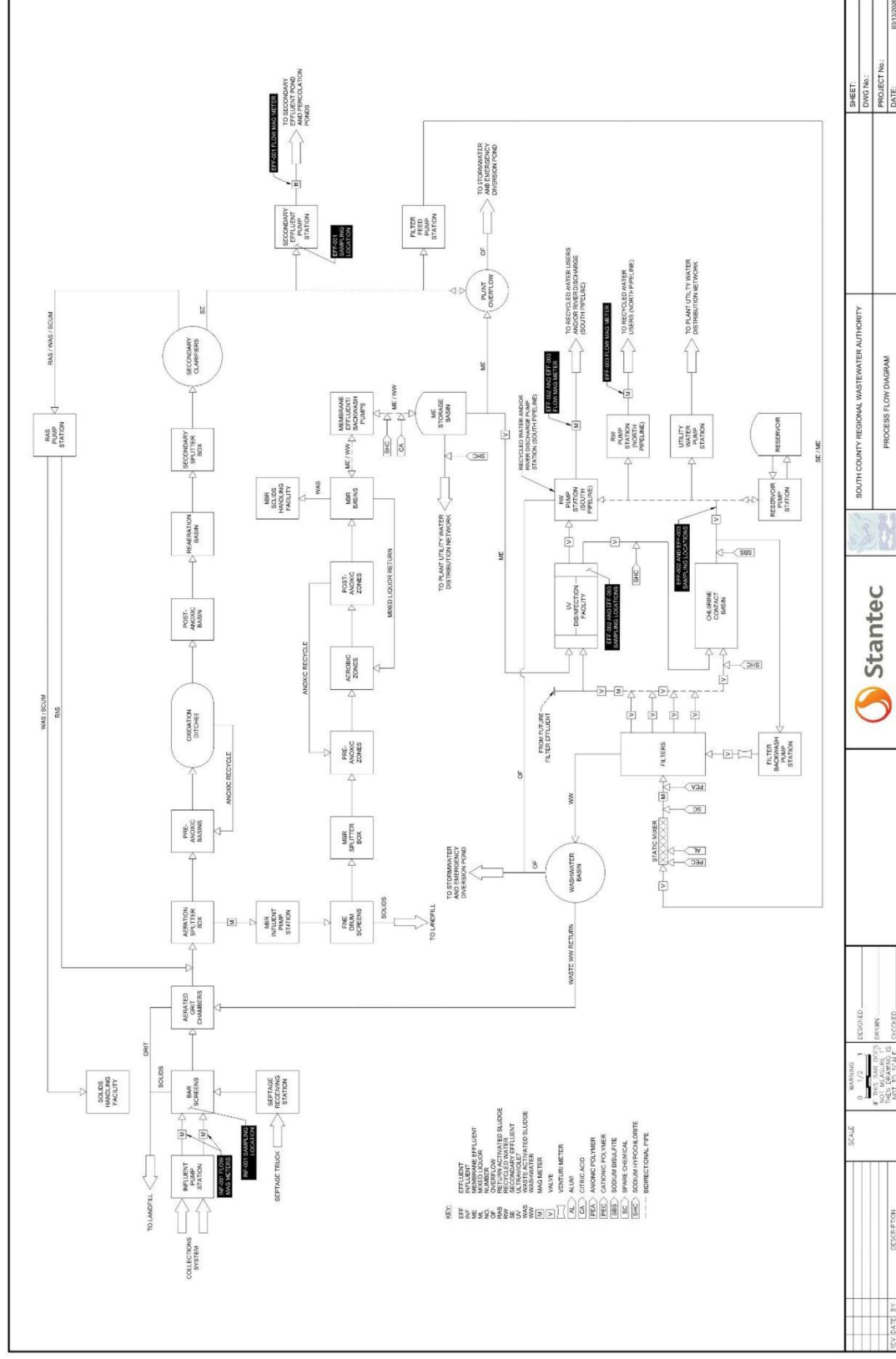
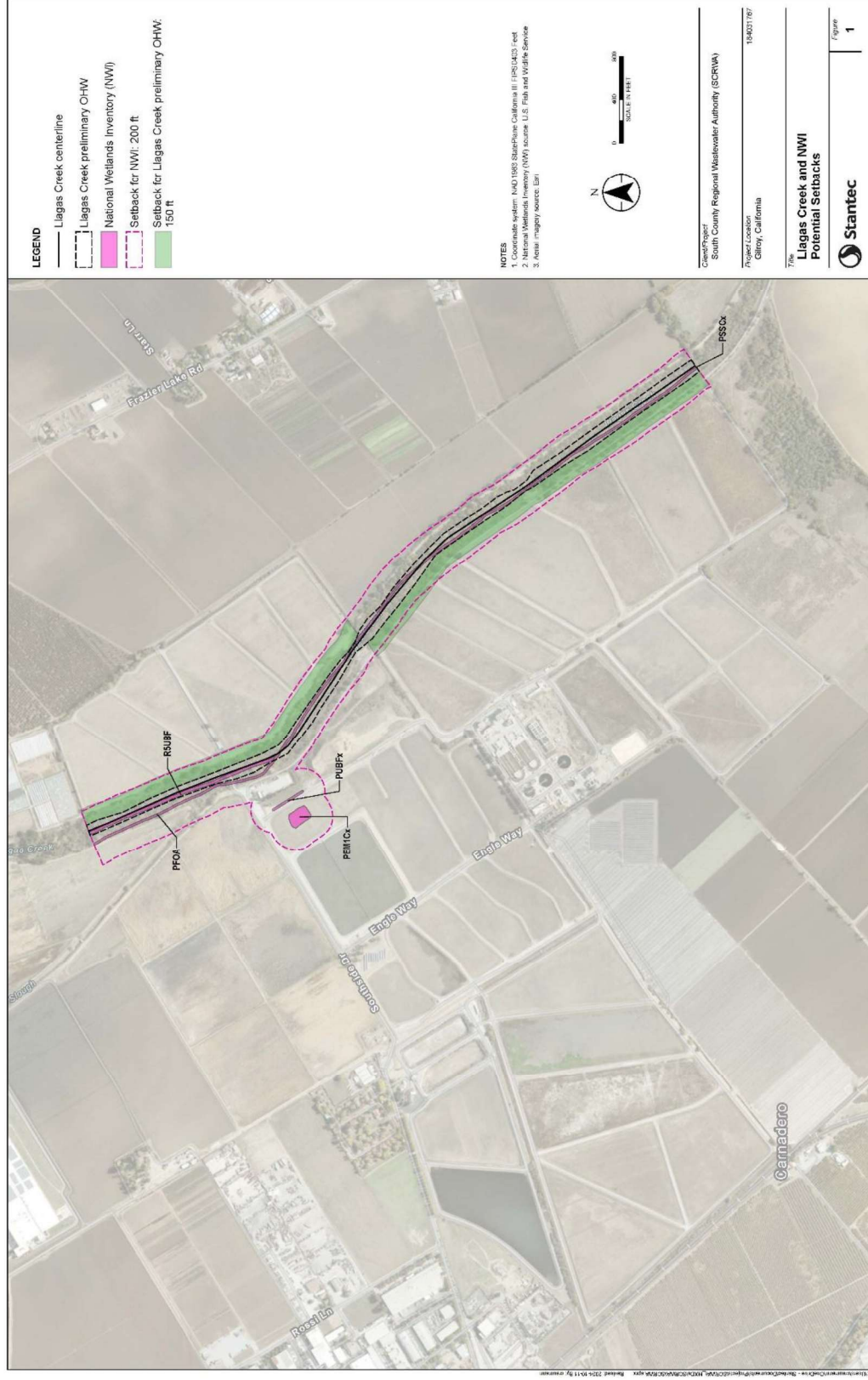


FIGURE 2. SITE PLAN



FIGURE 3. FACILITY MAP





**ATTACHMENT 3
MONITORING AND REPORTING PROGRAM**

**CENTRAL COAST REGIONAL WATER QUALITY CONTROL BOARD
MONITORING AND REPORTING PROGRAM R3-2026-0026
May 4, 2026**

**FOR
SOUTH COUNTY REGIONAL WASTEWATER TREATMENT AND RECLAMATION
FACILITY
GILROY, SANTA CLARA COUNTY**

This Monitoring and Reporting Program contains monitoring and reporting requirements for South County Regional Wastewater Treatment and Reclamation Facility (Wastewater System) enrolled in *General Waste Discharge Requirements Order R3-2020-0020 for Discharges from Domestic Wastewater Systems with Flows Greater than 100,000 Gallons per Day* (General Permit). This monitoring and reporting program is effective on July 1, 2026, concurrent with the effective dates of Order R3-2026-0014, a National Pollutant Discharge Elimination System permit authorizing disposal to the Pajaro River, and Order WQ 2016-0068-DDW, *Water Reclamation Requirements for Recycled Water Use*, and is contingent on Central Coast Regional Water Quality Control Board (Central Coast Water Board) adoption of the NPDES permit at its April 16, 2026 public meeting.

The South County Regional Wastewater Authority (SCRWA) owns and operates the Wastewater System that is subject to the General Permit and Notice of Applicability.

The State Water Resources Control Board (State Water Board) and Regional Water Quality Control Boards are transitioning to the use of the publicly accessible State Water Board's GeoTracker database for the tracking of environmental and regulatory data for sites that operate under waste discharge requirements. The monitoring and reporting program directs SCRWA to submit reports (both technical and monitoring reports) and analytical data electronically to the State Water Board GeoTracker database (see Electronic GeoTracker Submittal section below).

1. Sampling and Analysis

- 1.1. SCRWA must collect representative samples in accordance with the most recently approved sampling and analysis plan contained in the operations and maintenance manual required at section 10.3 of this MRP and validate analytical results prior to submittal to the Central Coast Water Board. All samples (e.g., wastewater, groundwater, soil, sludge) must be representative of the volume and nature of the discharge or matrix of materials sampled.
- 1.2. All samples must be collected by a qualified person, trained in proper procedures for collecting the samples. The name of the sampler, sample type (grab or composite), time, date, location, bottle/container type, and any preservative used for each sample must be recorded on the sample chain of custody form. The chain of custody form must also contain all custody information including date, time, and to whom samples were relinquished. If composite samples are collected, the basis for sampling (time or flow weighted) must be included in the sampling and analysis plan contained in the Operations and Maintenance Manual for Central Coast Water Board review and approval.
- 1.3. SCRWA may submit the same data to satisfy sampling requirements for the General Permit that is used to satisfy requirements in other orders as long as it meets all sampling criteria, including but not limited to method detection limit and reporting limit. Duplication of sampling is not intended.
- 1.4. Sampling Schedule

- 1.4.1. Unless otherwise specified below, sampling must be performed in accordance with Table 1.

Table 1. Sampling Schedule

Monitoring Period	Sample Collection Time
Monthly	Each month of the year
Quarterly	Quarter 1: January to March Quarter 2: April to June Quarter 3: July to September Quarter 4: October to December
Semiannual	Quarter 1 and Quarter 3
Annual	Quarter 3

- 1.4.2. Field test instruments (such as those used to test pH, temperature, dissolved oxygen, electrical conductivity, and settleable solids) may be

used if they are used by a State Water Board Environmental Laboratory Accreditation Program accredited laboratory, or:

- 1.4.2.1. The user is trained in proper use and maintenance of the instruments;
- 1.4.2.2. The instruments are field calibrated prior to monitoring events at the frequency recommended by the manufacturer;
- 1.4.2.3. Instruments are serviced and/or calibrated by the manufacturer at the recommended frequency; and
- 1.4.2.4. Field calibration reports are maintained and available for at least three years.

1.4.3. Monitoring Location Descriptions

- 1.4.3.1. All samples including influent samples (INF), effluent samples (EFF, FIL, UVS, CCB), and groundwater monitoring wells (GW) must be collected at the locations described in Table 2 and shown in Figure 2. These locations must be consistent with the Central Coast Water Board approved sampling and analysis plan. **SCRWA must upload the GeoTracker field point information for each sampling location in the GeoTracker database once, prior to uploading laboratory data (see Table 10).**

Table 2. Sample Identification Details^[1]

GeoTracker Field Point Class	GeoTracker Field Point Name (Sample ID) ^[2]	GeoTracker Field Point Description
Influent Sample	INF-001	Influent wastewater at the plant headworks, prior to treatment and following all significant input of wastewater to the treatment system
Effluent Sample	EFF-001	Secondary-treated wastewater sample collected prior to disposal to percolation ponds.
Effluent Sample	EFF-002	Disinfected tertiary-treated wastewater at a point after all treatment and prior to contact with the receiving water.
Effluent Sample	EFF-003	Disinfected tertiary-treated wastewater at a point after all treatment and prior to discharge to the recycled water distribution system
Effluent Sample	FIL-001	Granular Filter effluent
Effluent Sample	FIL-002	Membrane Bioreactor effluent

GeoTracker Field Point Class	GeoTracker Field Point Name (Sample ID) ^[2]	GeoTracker Field Point Description
Effluent Sample	UVS-001	Ultraviolet (UV) disinfection system
Effluent Sample	UVS-001A	Ultraviolet (UV) disinfection system channel 1
Effluent Sample	UVS-001B	Ultraviolet (UV) disinfection system channel 2
Effluent Sample	CCB-001	Sample for Chlorine contact basin
Remediation/ Groundwater Monitoring Well	GW-001	Groundwater well MW1/PWA located within the municipal ponds
Remediation/ Groundwater Monitoring Well	GW-002	Groundwater well PWF located within the municipal ponds
Remediation/ Groundwater Monitoring Well	GW-003	Groundwater well MW 13 located within the municipal ponds
Remediation/ Groundwater Monitoring Well	GW-004	Groundwater well MW 24 located within the municipal ponds
Remediation/ Groundwater Monitoring Well	GW-005	Groundwater well MW2/PWB located within the municipal ponds
Remediation/ Groundwater Monitoring Well	GW-006	Groundwater well MW7/PWT located within the municipal ponds
Remediation/ Groundwater Monitoring Well	GW-007	Groundwater well PWX located within the municipal ponds
Remediation/ Groundwater Monitoring Well	GW-008	Groundwater well PWC located within the municipal ponds
Remediation/ Groundwater Monitoring Well	GW-009	Groundwater well MW12/PWV located within the municipal ponds
Remediation/ Groundwater Monitoring Well	GW-010	Groundwater well PWY located within the municipal ponds
Remediation/ Groundwater Monitoring Well	GW-011	Groundwater well MW3/PWD located within the former food process ponds

GeoTracker Field Point Class	GeoTracker Field Point Name (Sample ID) ^[2]	GeoTracker Field Point Description
Remediation/ Groundwater Monitoring Well	GW-012	Groundwater well MW6/PWH located within the former food process ponds
Remediation/ Groundwater Monitoring Well	GW-013	Groundwater well MW4/PWK located within the former food process ponds
Remediation/ Groundwater Monitoring Well	GW-014	Groundwater well MW5/PWN located within the former food process ponds
Remediation/ Groundwater Monitoring Well	GW-015	Groundwater well PWR located within the former food process ponds
Remediation/ Groundwater Monitoring Well	GW-016	Groundwater well PWE located within the former food process ponds
Remediation/ Groundwater Monitoring Well	GW-017	Groundwater well PWI located within the former food process ponds
Remediation/ Groundwater Monitoring Well	GW-018	Groundwater well PWL located within the former food process ponds
Remediation/ Groundwater Monitoring Well	GW-019	Groundwater well PWP located within the former food process ponds
Remediation/ Groundwater Monitoring Well	GW-020	Groundwater well PWS located within the former food process ponds
Remediation/ Groundwater Monitoring Well	GW-021	Groundwater well MW8/PWG located within the former food process ponds
Remediation/ Groundwater Monitoring Well	GW-022	Groundwater well MW9/PWJ located within the former food process ponds
Remediation/ Groundwater Monitoring Well	GW-023	Groundwater well PWM located within the former food process ponds
Remediation/ Groundwater Monitoring Well	GW-024	Groundwater well PWQ located within the former food process ponds

GeoTracker Field Point Class	GeoTracker Field Point Name (Sample ID) ^[2]	GeoTracker Field Point Description
Remediation/ Groundwater Monitoring Well	GW-025	Groundwater well MW10 located north of Southside Drive
Remediation/ Groundwater Monitoring Well	GW-026	Groundwater well MW21 located east of the railway line
Remediation/ Groundwater Monitoring Well	GW-027	Groundwater well MW26 located within the municipal ponds
Stream or Creek Sampling Point	SW-009	Upstream surface water monitoring location in Llagas Creek approximately 1,000 feet north of Highway 152
Stream or Creek Sampling Point	SW-008	Upstream surface water monitoring location in Llagas Creek approximately 640 feet north of the northwest corner of Pond E1 and south of Highway 152
Stream or Creek Sampling Point	SW-001	Midstream surface water monitoring location at the drain on the south side of East Ponds at culvert prior to entering Llagas Creek
Stream or Creek Sampling Point	SW-005	Midstream surface water monitoring location at the outlet of Miller Slough near Luchessa Avenue bridge
Stream or Creek Sampling Point	SW-007	Downstream surface water monitoring location in Llagas Creek approximately 600 feet north of Bloomfield Road

[1] Water supply sample data should be included in the annual report but does not need to be uploaded to GeoTracker if the Central Coast Water Board approves submittal of a Well Identification Number pursuant to the Water Supply Monitoring section below.

[2] List does not include any new sampling field points identified in the groundwater sampling or surface water sampling plan. Data from any new sampling points identified in sampling plans must also be uploaded to GeoTracker.

2. Water Supply Monitoring

2.1. SCRWA must submit a Well Identification Number and the reporting year's consumer confidence report (annual water quality report or drinking water quality report), as required by the State Water Board Division of Drinking Water and/or county, provided, at a minimum, all the required constituents are sampled at the frequency specified in Table 3. SCRWA must report the results of any constituent monitored more frequently than is required by the monitoring program shown in Table 3. SCRWA must also report detectable concentrations (above the reporting limit) for any other constituent that has a published maximum contaminant level (MCL).

2.1.1. SCRWA must evaluate and provide a tabular summary of the water supply data with the annual monitoring report.

Table 3. Water Supply Monitoring

Constituent	Units	Sample Type	Sampling Frequency
Nitrate (as N)	mg/L ^[1]	Grab	Annually
Total Dissolved Solids	mg/L	Grab	Annually
Chloride	mg/L	Grab	Annually
Sodium	mg/L	Grab	Annually
Sulfate	mg/L	Grab	Annually
Boron	mg/L	Grab	Annually
Carbonate	mg/L	Grab	Annually
Bicarbonate	mg/L	Grab	Annually
Calcium	mg/L	Grab	Annually
Potassium	mg/L	Grab	Annually
Magnesium	mg/L	Grab	Annually

[1] mg/L denotes milligrams per liter

3. Influent and Effluent Monitoring

3.1. Influent and Effluent Flow Monitoring

3.1.1. SCRWA must monitor and report influent and effluent flow in million gallons per day, as described in Table 4, with each required monitoring report. For additional annual volumetric reporting requirements, see the State Water Board Volumetric Annual Reporting section of this Monitoring and Reporting Program.

Table 4. Influent and Effluent Flow Monitoring

Parameter	Units	Monitoring Type	Monitoring Frequency	Reporting Requirements
Influent Flow	MGD ^[1]	Metered / Calculated	Daily	Daily Flow, Total Average Monthly ^[2]
Effluent Flow to Ponds (EFF-001)	MGD	Metered / Calculated	Daily	Daily Flow, Total Average Monthly
Recycled Water Flow (EFF-003)	MGD	Metered / Calculated	Daily	Daily Flow, Maximum Daily, Total Average Monthly
Flow at Chlorine Contact Basin (CCB-001)	MGD	Calculated	Continuously	Maximum Daily
Flow at each UV Channel (UVS-001A and UVS-001B)	MGD	Calculated	Continuously	Maximum Daily Flow
Total Average Monthly Flow (combined EFF-001 and EFF-003)	MGD	Calculated	Monthly	Monthly Average
Percent of Permitted Flow ^[3]	MGD	Calculated	Monthly	Monthly Percentage

[1] denotes million gallons per day.

[2] Average monthly to be calculated by the total flow by volume during a calendar month divided by the number of days in the month that the system received flow.

[3] The General Permit Notice of Applicability specifies the Wastewater System's maximum permitted dry-weather flow. In the monitoring reports, SCRWA must evaluate and provide a comparison of monitored effluent flow at EFF-001, EFF-002, and EFF-003 to the permitted average dry-weather flow.

3.2. Wastewater System Monitoring

3.2.1. Representative samples of SCRWA's influent (raw wastewater) into the Wastewater System and effluent (treated wastewater) discharged to land and recycled must be collected and analyzed in accordance with the treatment technology-based monitoring requirements (including sample type and frequency) summarized in the tables below. See Figure 1 for the location of samples and GeoTracker field points.

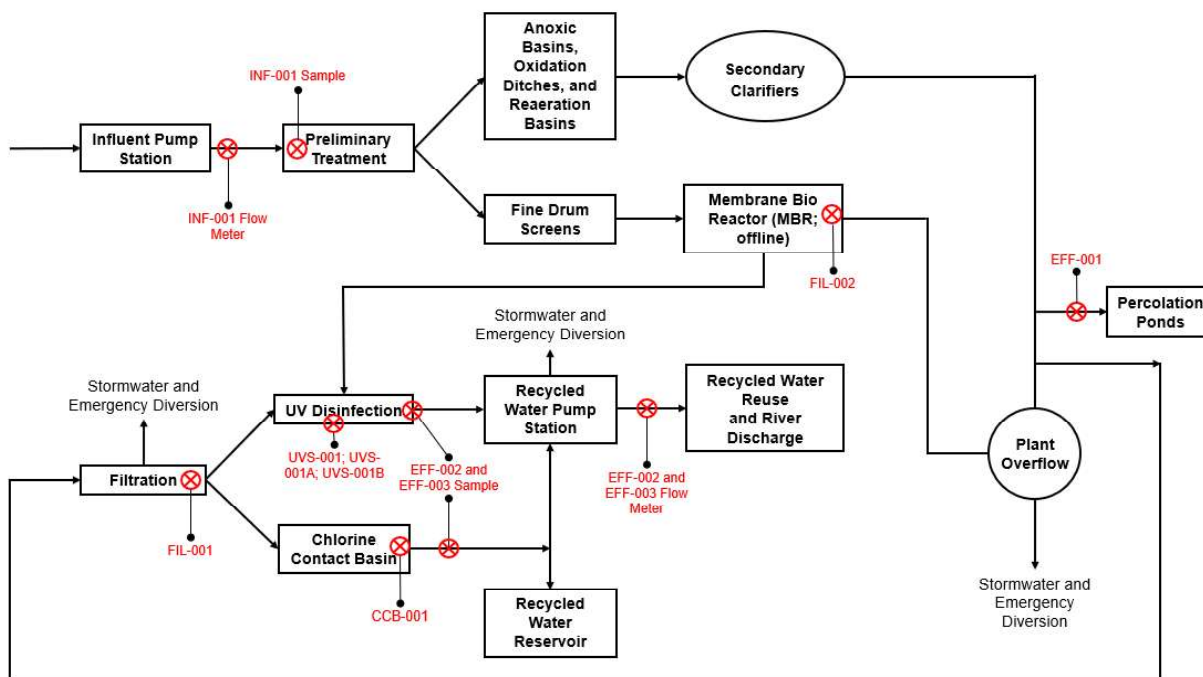


Figure 1. Simplified Block Diagram and Sample Locations

3.3. Influent and Effluent Monitoring for Enhanced Activated Sludge Wastewater Systems

- 3.3.1. At a minimum, influent and effluent constituent monitoring must be consistent with Table 5.
- 3.3.2. SCRWA must calculate 7-day averages, 30-day averages, 25-month rolling medians, and daily and monthly maximums for select analytes when comparing monitoring data to applicable limitations specified in the General Permit.

Table 5. Influent and Effluent Monitoring for Enhanced Activated Sludge and Tertiary Treatment

Constituent	Units	INFLUENT Sample Type	INFLUENT Sampling Frequency	EFFLUENT Sample Type	EFFLUENT Sampling Frequency	EFFLUENT Sample Location
pH	units	Grab	Weekly	Grab	Weekly	EFF-001
Biochemical Oxygen Demand, 5-Day	mg/L ^[1]	24-hour composite	Quarterly	24-hour composite	Monthly	EFF-001 and EFF-003
Total Suspended Solids	mg/L	24-hour composite	Quarterly	24-hour composite	Monthly	EFF-001 and EFF-003

Constituent	Units	INFLUENT Sample Type	INFLUENT Sampling Frequency	EFFLUENT Sample Type	EFFLUENT Sampling Frequency	EFFLUENT Sample Location
Settleable Solids	ml/L ^[2]	Not Required	Not Required	Grab	Weekly	EFF-001
Total Nitrogen ^[3]	mg/L	Calculated	Quarterly	Calculated	Monthly	EFF-001 and EFF- 003
Nitrate (as N)	mg/L	24-hour composite	Quarterly	Grab	Monthly	EFF-001
Nitrite (as N)	mg/L	24-hour composite	Quarterly	Grab	Monthly	EFF-001
Total Kjeldahl Nitrogen (as N)	mg/L	24-hour composite	Quarterly	Grab	Monthly	EFF-001
Un-ionized Ammonia (as N) ^[4]	mg/L	24-hour composite	Quarterly	Grab	Monthly	EFF-001
Ammonia (as N)	mg/L	24-hour composite	Quarterly	Grab	Monthly	EFF-001
Total Dissolved Solids	mg/L	24-hour composite	Quarterly	Grab	Quarterly	EFF-001
Chloride	mg/L	24-hour composite	Quarterly	Grab	Quarterly	EFF-001
Sodium	mg/L	24-hour composite	Quarterly	Grab	Quarterly	EFF-001
Sulfate	mg/L	24-hour composite	Quarterly	Grab	Quarterly	EFF-001
Boron	mg/L	24-hour composite	Quarterly	Grab	Quarterly	EFF-001
Carbonate	mg/L	Not Required	Not Required	Grab	Semiannually	EFF-001
Bicarbonate	mg/L	Not Required	Not Required	Grab	Semiannually	EFF-001
Calcium	mg/L	Not Required	Not Required	Grab	Semiannually	EFF-001
Potassium	mg/L	Not Required	Not Required	Grab	Semiannually	EFF-001
Magnesium	mg/L	Not Required	Not Required	Grab	Semiannually	EFF-001
Zinc	mg/L	Grab	Quarterly	Grab	Semiannually	EFF-001

[1] mg/L denotes milligrams per liter

[2] mL/L denotes milliliters per liter

[3] Total nitrogen is the sum of total inorganic nitrogen (nitrate + nitrite + ammonium + ammonia) and organic nitrogen.

- [4] Temperature and pH are to be measured at the same time as the total ammonia sample is collected. Results shall be used to calculate and report unionized ammonia concentrations.

3.4. Effluent Monitoring for Non-Potable Recycled Water Producers

3.4.1. SCRWA is required to comply with the non-potable recycled water monitoring requirements specified in Table 6 and the Division of Drinking Water-approved Title 22 engineering report. If there is an overlap in constituent-specific monitoring, the Title 22 engineering report governs.

3.4.2. SCRWA must maintain continuous monitoring data onsite and only submit the daily maximum, minimum, or average values, as described in the “Data Submittal Requirement” column. All underlying continuous data must be retained and made available upon request.

Table 6. Monitoring for Non-Potable Recycled Water

Constituent or System Operation Parameter	Units	Sample Type	Sampling Frequency	Reporting Requirement	Sampling Location
Recycled Water Produced	Acre-feet	Calculated	Monthly	Monthly Total	EFF-003
Recycled Water Distributed	Acre-feet	Calculated	Monthly	Monthly Total	EFF-003
Total Coliform ^[1]	MPN/100 mL ^[2]	Grab	Daily	Daily, Seven Day Median, Maximum Monthly	EFF-003
Turbidity ^[1]	NTU ^[3]	Metered	Continuously ^[4]	Average Daily, Maximum Daily,	FIL-001 & FIL-002
Chlorine Residual ^[1]	mg/L ^[5]	Metered	Continuously	Average Daily, Minimum Daily,	CCB-001
UV Transmittance (UVT)	Percent (%)	Metered	Continuously	Average Daily, Maximum Daily, Minimum Daily,	UVS-001
UV Dose	mJ/cm ²	Metered	Continuously	Minimum Daily, Average Daily,	UVS-001

Constituent or System Operation Parameter	Units	Sample Type	Sampling Frequency	Reporting Requirement	Sampling Location
UV Sensor Intensity	mW/cm ²	Metered	Continuously	Average Daily, Maximum Daily, Minimum Daily,	UVS-001

- [1] Report values as required by the most recent conditionally accepted Title 22 engineering report and/or Division of Drinking Water letter.
- [2] MPN/100 mL denotes most probable number per 100 mL sample.
- [3] NTU denotes nephelometric turbidity unit.
- [4] Continuous monitoring for the filtration and disinfection (UV) system must be conducted following approved standard operating procedures in the operation and maintenance manual.
- [5] denotes milligrams per liter

4. Disposal Monitoring

- 4.1. SCRWA must monitor all ponds (SH1, SH2, Ponds C1 through C4, Secondary Effluent Pond, Ponds D1 through D6, Ponds S1 through S9, Ponds ND 8 through ND17, Ponds E1 through E7, and the Area 98 Reservoir) when wastewater is applied. Inspections (observations) must be conducted as described in Table 7. Notations must be made in a bound logbook, and problems must be promptly corrected and recorded. A log of these inspections must be maintained onsite and made available to Central Coast Water Board upon request.
- 4.2. Observations do not need to be reported in the monitoring reports unless a violation, significant change, or treatment issue is observed.
- 4.3. Photos of the pond(s) verifying the observations must be collected annually and included in the annual report, with the date and location of the photo clearly marked.
- 4.4. SCRWA must evaluate and summarize wastewater disposal application rates, loading rates (pounds/acre/day), violations found during the inspections, with each monitoring report.
- 4.5. If treated effluent is not discharged to land during a reporting period, the monitoring report must still be submitted and indicate that there was no discharge during the reporting period.

Table 7. Effluent Disposal Monitoring

Constituent	Units	Sample Type	Monitoring Frequency
Freeboard	0.1 feet	Measured	Weekly

Constituent	Units	Sample Type	Monitoring Frequency
Supplemental Irrigation	Gallons per day	Metered/Estimated ^[1]	Daily
Local Precipitation	Inches/day	Weather Station ^[2]	Each precipitation event ^[3]
Acreage Applied ^[4]	Acres	Measured	Daily
Application Rate	Gallons per day	Metered/Estimated ^[1]	Daily
Average Monthly Biochemical Oxygen Demand, 5-Day (BOD) Applied ^[5] ^[6]	lbs/acre/day	Calculated	Monthly
Total Nitrogen Applied ^[5] ^[6]	lbs/acre/day	Calculated	Monthly
Salts Applied (total dissolved solids, sodium, chloride, sulfate, boron) ^[5] ^[6]	lbs/acre/day	Calculated	Monthly
Soil Erosion Evidence	Not Applicable	Observation	Monthly
Vegetation in and Surrounding Ponds	Not Applicable	Observation	Monthly
Containment Berm Condition	Not Applicable	Observation	Monthly
Soil Saturation/Ponding	Not Applicable	Observation	Monthly
Nuisance Odors/Vectors	Not Applicable	Observation	Monthly
Discharge Offsite	Not Applicable	Observation	Monthly
Sludge Depth	0.1 feet	Measured	Annually

[1] Requires meter reading, a pump run time meter, or other approved method. If the flow is estimated, SCRWA is required to provide an explanation (e.g., no meter- estimated using pump operations including rate and time) with each monitoring report.

- [2] SCRWA may use a rain gauge or use a National Oceanic and Atmospheric Administration, United States Geological Survey, or California Irrigation Management Information System weather station, such as <http://scacis.rcc-acis.org/>.
- [3] A precipitation event is one producing precipitation of ½ inch or more.
- [4] Acreage applied denotes the acreage to which wastewater is applied.
- [5] Applied loading rates are calculated as described in Land Application Mass Loading Calculation section of this Monitoring and Reporting Program
- [6] Frequencies of analytical testing are defined in the influent and effluent monitoring tables.

4.6. Land Application Mass Loading Calculation

Land Application Mass Loading Calculation (lb/acre/day)

Total Nitrogen/Salts/BOD Applied (pounds/acre/ day) =

$$\frac{X \text{ [mg/L]} \times Q \text{ [million gallons per day]} \times 8.34}{\text{Acreage Applied [acres]}}$$

Where X = Total nitrogen, salts, or BOD concentration

Where Q = Application Rate (often effluent flow rate)

Where 8.34 = Unit conversion factor from mg/L to pounds/million gallons per day

5. Sludge/Biosolids Disposal Monitoring

- 5.1. SCRWA must report the handling and disposal of all sludge/biosolids generated by the wastewater system in accordance with the requirements of the current NPDES Permit.

6. Groundwater Monitoring

- 6.1. Reported water quality data indicate that concentrations of total dissolved solids, chloride, sodium, sulfate, and boron in effluent exceed the General Permit effluent limitations for the Pajaro Valley Basin Llagas Sub-Area. Therefore, in accordance with General Permit section V.A.2.ii, SCRWA's compliance point for compliance and monitoring of these constituents is groundwater (Option 2). SCRWA must therefore implement a groundwater monitoring program to demonstrate compliance with the groundwater limitations in the Notice of Applicability and water quality objectives specified in the Basin Plan.
- 6.2. The current groundwater monitoring network consists of 27 monitoring wells that are between 19.4 feet to 300 feet below ground surface (for wells with known depths). The current upgradient monitoring well has a total depth of approximately 300 feet below ground surface and the screened interval is unknown. SCRWA must sample the existing groundwater monitoring network GW-001 through GW-027 until an updated groundwater monitoring work plan is approved by the Central Coast Water Board.

6.3. **Within six months** of the effective date of the Notice of Applicability, SCRWA must:

- 6.3.1. Submit a groundwater monitoring work plan and preliminary hydrogeologic conceptual site model for Central Coast Water Board approval. At a minimum, the work plan must evaluate the existing monitoring well network and, if necessary, propose the installation or identification of additional groundwater monitoring wells necessary to better characterize groundwater gradients, groundwater flow velocity, and direction to evaluate any influence of effluent disposal on groundwater quality and surface water volume and quality. The work plan must also evaluate if any existing wells need to be rehabilitated or replaced. The number and locations of the proposed groundwater monitoring wells (existing, new, and/or redrilled/rehabilitated) should be adequate to meet the goals of the work plan as noted above. The proposed groundwater monitoring well network must be on a scale sufficient to determine potential impacts and be sufficiently representative of groundwater conditions upgradient and downgradient of the permitted disposal/dispersal area. If appropriate, SCRWA must evaluate if wells are necessary to monitor vertical gradients and to characterize groundwater water quality between shallow and deep aquifers. The groundwater monitoring wells should not be installed prior to Central Coast Water Board approval of the groundwater monitoring work plan and preliminary hydrogeologic conceptual site model. Groundwater monitoring workplans, and other technical reports, that interpret groundwater data must be prepared and stamped by a California licensed civil engineer or professional geologist
- 6.3.2. Any new groundwater monitoring wells must be installed in accordance with California Department of Water Resources Bulletin 74-81 and Supplement 74-90, California Water Code sections 13710 through 13755, and any local permitting requirements. The locations and top-of-casing elevations for the newly installed monitoring wells must be surveyed by a licensed land surveyor. SCRWA must install⁷ the Central Coast Water Board-approved groundwater monitoring well network (new and/or rehabilitated/replaced) within **one year** of receiving Central Coast Water Board approval.
- 6.3.3. All new and existing monitoring wells (shown in Table 2 of this Monitoring and Reporting Program) must be, at a minimum, sampled and analyzed as specified in Table 8 and the Central Coast Water Board-approved sampling and analysis plan (see section VI.A.2.i of the General Permit).
- 6.3.4. Prior to sampling, the Discharger must measure depth to groundwater and calculate groundwater elevations.⁸ The monitoring wells must be purged of

⁷ Well installation must be supervised by a California-licensed civil engineer or professional geologist.

⁸ The locations and top-of-casing elevations for the existing groundwater monitoring wells must be surveyed by a licensed land surveyor if not already completed at the time of installation.

a minimum of at least three well volumes. Well purging must continue until measurements of the following parameters have stabilized (i.e., are reproducible within 10 percent): pH, temperature, dissolved oxygen, electrical conductivity, and turbidity. No-purge, low-flow, or other sampling techniques are acceptable only if they are approved in advance by the Central Coast Water Board and described in an approved sampling and analysis plan. Once the groundwater level in each of the wells has recovered sufficiently to ensure the collection of representative groundwater samples, a qualified individual (e.g., consultant, technician) trained in using proper sampling methods must recover samples using approved USEPA methods. Laboratories analyzing groundwater samples must be accredited by the State Water Board Environmental Laboratory Accreditation Program, in accordance with California Water Code section 13176, and must include quality assurance/quality control data with their reports.

6.3.5. SCRWA must provide monitoring well field sheets and report monitoring data, as described in Table 8, with each monitoring report.

6.3.6. Groundwater monitoring, at a minimum, must comply with Table 8.

Table 8. Groundwater Monitoring

Constituent	Units	Sample Type	Sampling Frequency
Groundwater Elevation ^[1]	0.01 ft msl ^[2]	Calculated	Quarterly
Depth to Groundwater ^[1]	0.01 ft bgs ^[2]	Measurement	Quarterly
Gradient	feet/feet	Calculated	Quarterly
Groundwater Flow Direction ^[1]	degrees	Calculated	Quarterly
pH	pH Units	Metered	Quarterly
Dissolved Oxygen	mg/L ^[3]	Metered	Quarterly
Electrical Conductivity	µmhos/cm ^[4]	Metered	Quarterly
Oxidation Reduction Potential	mV ^[5]	Metered	Quarterly
Temperature	degrees Celsius	Metered	Quarterly
Total Nitrogen ^[6]	mg/L	Calculated	Quarterly
Nitrate (as N)	mg/L	Grab	Quarterly
Nitrite (as N)	mg/L	Grab	Quarterly
Total Kjeldahl Nitrogen (as N)	mg/L	Grab	Quarterly

Constituent	Units	Sample Type	Sampling Frequency
Ammonia (as N)	mg/L	Grab	Quarterly
Total Dissolved Solids	mg/L	Grab	Quarterly
Chloride	mg/L	Grab	Quarterly
Sodium	mg/L	Grab	Quarterly
Sulfate	mg/L	Grab	Quarterly
Boron	mg/L	Grab	Quarterly
Carbonate	mg/L	Grab	Semiannually
Bicarbonate	mg/L	Grab	Semiannually
Calcium	mg/L	Grab	Semiannually
Potassium	mg/L	Grab	Semiannually
Magnesium	mg/L	Grab	Semiannually
Zinc ^[7]	mg/L	Grab	Quarterly

[1] Calculations must be prepared by, or under the responsible charge of, a California licensed professional.

[2] ft msl denotes feet above mean sea level and ft bgs denotes feet below ground surface.

[3] mg/L denotes milligrams per liter

[4] μ mhos/cm denotes micromhos per centimeter

[5] mV denotes millivolts

[6] Total nitrogen is the sum of total inorganic nitrogen (nitrate + nitrite + ammonium + ammonia) and organic nitrogen.

[7] Sampling may be reduced to annual if concentrations are not detected after eight quarters.

7. Surface Water Monitoring

7.1. Treated effluent discharged to land may have the potential to impact surface water quality. Due to the disposal ponds' proximity to Llagas Creek, and pursuant to General Permit section II.B.2, SCRWA must monitor Llagas Creek at a minimum as described in Table 9.

7.2. **Within 120 days** of issuance of the Notice of Applicability, SCRWA must:

7.2.1. Submit a surface water monitoring work plan for Central Coast Water Board approval. The number and location of the surface water monitoring locations should be adequate to characterize surface water quality upstream, midstream, and downstream of the percolation ponds and, if necessary, water quality from tributary streams to Llagas Creek. At a minimum, the workplan must evaluate existing surface water monitoring locations and, if necessary, propose additional surface water sampling.

7.3. After eight quarters of data, SCRWA may request to discontinue or reduce surface water sampling for all or a subset of constituents if the data shows disposal of treated effluent is not impacting Llagas Creek for those constituents. This reduction request must be approved by the Central Coast Water Board.

Table 9. Surface Water Monitoring

Constituent	Units	Sample Type	Sampling Frequency
Flow Rate	cubic feet per second	Measured	Quarterly
pH	pH Units	Metered	Quarterly
Dissolved Oxygen	mg/L ^[1]	Metered	Quarterly
Electrical Conductivity	µmhos/cm ^[2]	Metered	Quarterly
Oxidation Reduction Potential	mV ^[3]	Metered	Quarterly
Temperature	degrees Celsius	Metered	Quarterly
Total Nitrogen ^[4]	mg/L	Calculated	Quarterly
Nitrate (as N)	mg/L	Grab	Quarterly
Nitrite (as N)	mg/L	Grab	Quarterly
Total Kjeldahl Nitrogen (as N)	mg/L	Grab	Quarterly
Ammonia (as N)	mg/L	Grab	Quarterly
Total Dissolved Solids	mg/L	Grab	Quarterly
Chloride	mg/L	Grab	Quarterly
Sodium	mg/L	Grab	Quarterly
Sulfate	mg/L	Grab	Quarterly
Boron	mg/L	Grab	Quarterly
Carbonate	mg/L	Grab	Semiannually
Bicarbonate	mg/L	Grab	Semiannually
Calcium	mg/L	Grab	Semiannually
Potassium	mg/L	Grab	Semiannually
Magnesium	mg/L	Grab	Semiannually

Constituent	Units	Sample Type	Sampling Frequency
Zinc	mg/L	Grab	Quarterly

[1] mg/L denotes milligrams per liter.

[2] μ mhos/cm denotes micromhos per centimeter

[3] mV denotes millivolts

[4] Total nitrogen is the sum of total inorganic nitrogen (nitrate + nitrite + ammonium + ammonia) and organic nitrogen.

8. Reporting Requirements

8.1. Quarterly and Annual Monitoring Reports

8.1.1. Quarterly and annual monitoring reports must be submitted in accordance with the schedule shown in Table 10.

Table 10. Quarterly and Annual Monitoring Reporting Requirements

Report	Monitoring Period	Report Due Date
First Quarter Monitoring Report	January 1 to March 31	May 1
Second Quarter Monitoring Report	April 1 to June 30	August 1
Third Quarter Monitoring Report	July 1 to September 30	November 1
Fourth Quarter Monitoring Report	October 1 to December 31	February 1
Annual Report	January 1 to December 31	April 30
State Water Board Volumetric Annual Reporting	January 1 to December 31	April 30

8.2. Quarterly Monitoring Reporting

8.2.1. At a minimum, the quarterly reports must include the following:

8.2.1.1. Results of all required monitoring presented in tabular format.

8.2.1.2. Results of any pollutant or parameter monitored more frequently than required by this Monitoring and Reporting Program. Data obtained through additional monitoring must be used in calculations, as appropriate.

8.2.1.3. A comparison of monitoring data to the discharge specifications, applicable effluent limitations, disclosure of any violations of the

Notice of Applicability and/or General Permit, and an explanation of any violation of those requirements. **SCRWA must calculate 7-day averages, 30-day averages, 25-month rolling medians, and daily and monthly maximums for select analytes when comparing monitoring data to applicable limitations specified in the General Permit.** Data must be presented in tabular format.

- 8.2.1.4. A table showing groundwater well construction details including: well number or ID, groundwater zone, coordinates (longitude and latitude), ground surface elevation, reference elevation, elevation of screen(s), elevation of bentonite, elevation of filter pack, and elevation of well bottom.
- 8.2.1.5. Cumulative data tables containing: depth to groundwater, groundwater elevation, and water quality analytical results. This table should compare groundwater quality data to the groundwater limitations and identify if the wells are upgradient, cross-gradient, or downgradient of the Wastewater System.
- 8.2.1.6. Copies of laboratory analytical report(s) and chain of custody form(s).
- 8.2.1.7. Copies of groundwater monitoring well field sheets with purge methods and data.

8.3. Annual Reporting

8.3.1. Annual Report Template

- 8.3.1.1. SCRWA must submit annual reports in compliance with Standard Provisions 2013⁹ (and any updates to the Standard Provisions) Section C, General Reporting Requirements, Item 16 using the most recent annual report template provided by Central Coast Water Board. The current annual report template can be found here:

https://www.waterboards.ca.gov/centralcoast/board_decisions/adopted_orders/2020/wdr_annual_report_format.pdf

8.3.2. Groundwater Monitoring and Reporting

- 8.3.2.1. SCRWA must include a section in its annual report titled Groundwater Monitoring and Reporting that evaluates groundwater conditions in the vicinity of the Wastewater System. SCRWA must include:
 - 8.3.2.1.1. An evaluation of groundwater flow and overall impacts of effluent on groundwater quality and quantity, and any new

⁹ See Standard Provisions, 2013:

https://www.waterboards.ca.gov/centralcoast/board_decisions/docs/wdr_standard_provisions_2013.pdf

information generated from the monitoring well boring logs, and groundwater elevation data.

- 8.3.2.1.2. Field logs that contain, at a minimum, sampling method, water quality parameters measured during purging (if purging is necessary for the sample method) or sampling, method of purging, depth to water, volume of water purged, etc.
- 8.3.2.1.3. Groundwater contour maps for the reporting period for all groundwater zones, if applicable.
- 8.3.2.1.4. A site map showing the location of wells and their concentrations total dissolved solids, chloride, sodium, sulfate and boron in groundwater.
- 8.3.2.1.5. A table showing groundwater well construction details such as well number, groundwater zone, coordinates (longitude and latitude), ground surface elevation, reference elevation, elevation of screen, elevation of bentonite, elevation of filter pack, and elevation of well bottom.
- 8.3.2.1.6. A table showing historical lateral and vertical (if applicable) flow directions and gradients.
- 8.3.2.1.7. Cumulative data tables containing the groundwater quality analytical results and depth to groundwater. This table must compare groundwater water quality data to the groundwater limitations and to the effluent concentrations. The table should additionally identify wells as up-gradient, cross-gradient, or down-gradient of the Wastewater System.
- 8.3.2.1.8. Results of any monitoring done more frequently than required at the locations specific in this General Permit.
- 8.3.2.1.9. A trend analysis that summarizes if the concentrations of total dissolved solids, chloride, sodium, sulfate, and boron are stable, increasing, or decreasing and their degree of significance. The cause for a statistically significant increasing trend must be discussed if any statistically significant increasing trend is observed.
- 8.3.2.1.10. Identification of data gaps and potential deficiencies in the groundwater monitoring system or reporting program.
- 8.3.2.1.11. As required by the California Business and Professions Code sections 6735, 7835, and 7835.1, all groundwater monitoring reports must be prepared under the supervision of a registered professional engineer or geologist and signed by the registered professional.

8.3.3. Maintenance Activities

- 8.3.3.1. A statement that the minimum maintenance was performed in accordance with the Operations and Maintenance Manual and Computerized Maintenance Management System (CMMS). Maintenance work orders must be available for review upon request.

8.4. State Water Board Volumetric Annual Reporting

- 8.4.1. SCRWA must electronically certify and submit a summary of the monthly influent and effluent flow volume by April 30 of each calendar year via the State Water Board's Internet GeoTracker database.¹⁰ The submittal must include:
 - 8.4.1.1. Monthly volume of wastewater collected and treated by the wastewater treatment plant (i.e. total monthly influent volume).
 - 8.4.1.2. Monthly volume of wastewater treated, specifying level of treatment, including treated wastewater discharged.
 - 8.4.1.3. If applicable, monthly volume of recycled water distributed, and annual volume of treated wastewater distributed for beneficial use in compliance with California Code of Regulations, title 22 use categories.
- 8.4.2. Additional information about volumetric annual reporting of wastewater and recycled water and the Recycled Water Policy is available at the Recycled Water Policy Volumetric Annual Reporting Website.¹¹

8.5. Non-Compliance Reporting:

- 8.5.1. Pursuant to General Permit section VI.C.2, SCRWA must notify and report to Central Coast Water Board non-compliance of limits related to pond freeboard, flow rate, bypass or overflow, delivery of off-specification recycled water, and wastewater containment failure. Additionally, SCRWA must notify and report to Central Coast Water Board any non-compliance with applicable Title 22 requirements, site-specific requirements found in its conditional acceptance letter, or the approved Title 22 engineering report.

9. Electronic GeoTracker Submittal

- 9.1. All monitoring reports must be provided electronically in a searchable PDF format, with the Central Coast Water Board's current transmittal sheet found at the link below as the cover page. The transmittal sheet must be signed.

https://www.waterboards.ca.gov/centralcoast/water_issues/programs/wastewater_permitting/docs/transmittal_sheet.pdf

¹⁰ See: <https://geotracker.waterboards.ca.gov/>

¹¹ See: [Volumetric Annual Reporting | California State Water Resources Control Board](#)

- 9.2. The transmittal sheet identified authorized signatories. To authorize a representative to sign and submit reports on SCRWA's behalf, SCRWA must submit a "designation of duly authorized representatives form" to the Central Coast Water Board.

https://www.waterboards.ca.gov/centralcoast/water_issues/programs/wastewater_permitting/

- 9.3. SCRWA must submit all reports/documents and laboratory analytical data to the State Water Board's GeoTracker^{12,13} database consistent with applicable Electronic Submittal of Information (ESI) requirements under the Wastewater System-specific global identification number **WDR100034821** at:

<https://geotracker.waterboards.ca.gov/esi/login>

- 9.4. For general questions, please contact the GeoTracker Help Desk at:
Geotracker@waterboards.ca.gov.

- 9.5. Table 11 summarizes all the GeoTracker electronic reporting requirements. Central Coast Water Board may request submittal of some documents on paper, particularly drawings or maps that require a large size to be readable, or in other electronic formats where evaluation of data is required.

Table 11. GeoTracker Electronic Submittal Information Data Requirements

Electronic Submittal	Description of Action	Action	Frequency
Reports and Documents	Complete copy of all documents including monitoring reports (in searchable PDF format) and any other associated documents related to the Wastewater System.	Upload directly to GeoTracker all monitoring reports (in searchable PDF format) and any other associated documents.	On or before the due dates required by this General Permit and for other documents when required by the Central Coast Water Board.

¹² Information for first-time GeoTracker users is available at:

https://www.waterboards.ca.gov/ust/electronic_submittal/docs/beginnerguide2.pdf

¹³ Additional information available at:

<https://geotracker.waterboards.ca.gov/>

Electronic Submittal	Description of Action	Action	Frequency
Laboratory Data	All analytical data (including geochemical data) in electronic deliverable format (EDF). This includes all water, soil, and vapor samples collected when monitoring a discharge.	Upload or direct your California ELAP-accredited laboratory staff to upload all EDF laboratory data directly to GeoTracker.	On or before the due date of the required monitoring report.
Depth to Groundwater	Monitoring wells must have the depth-to-water information reported. Report data only for wells defined as permanent sampling points.	Upload depth-to-water information to the GeoTracker GEO_WELL file.	On or before the due date of the required monitoring report.
Boring Logs and Well Screen Intervals	Boring logs must be prepared by a registered professional and submitted in PDF format separately (not only as attachments to reports).	Upload boring logs (in searchable PDF format) to GeoTracker GEO_BORE file whenever a new boring is drilled.	Every time a new boring is drilled.
Field Points, Location Data (Geo XY) ^[1]	Name, classify, and identify the location (latitude and longitude) of all sampling points. Monitoring wells must be surveyed, influent and effluent sample locations must be identified on the GeoTracker mapping tool under “non-surveyed data.” These data points are required prior to laboratory data uploads.	Upload the location data (surveyed and non-surveyed) to the GeoTracker Geo_XY file.	Every time a permanent monitoring point is established.

Electronic Submittal	Description of Action	Action	Frequency
Elevation Data (Geo Z) ^[2]	Survey and mark the elevation at the top of groundwater well casings for all permanent groundwater wells. These points are required prior to depth-to-water data uploads.	Upload the survey data to the GeoTracker GEO_Z file.	One-time, for all groundwater monitoring wells.
Geo Map	Site layout, map of facilities, Wastewater System including treatment and disposal area(s).	Upload the Site layout PDF to the GeoTracker site plan file.	Year one and every five years thereafter and when the facilities are modified.

[1] Geo XY required for all wells. New wells must be surveyed. For existing wells, use original well installation survey data. SCRWA must also upload sample locations (e.g., influent and effluent samples) that are not defined as a **permanent monitoring well** and have not been surveyed by a licensed professional.

[2] Geo Z required for all wells. New wells must be surveyed. For existing wells, use original well installation survey data.

10. Technical Reports and One-Time Submittals

10.1. The technical reports are due as described in Table 12. The General Permit typically requires a technical report for a pretreatment program and a sludge management plan. These activities are not covered under this Notice of Applicability as SCRWA's pretreatment and biosolids programs are regulated pursuant to the NPDES permit.

Table 12. Technical Report Submittal Due Dates

Report	Report Due Date ^[1]
Final Title 22 Engineering Report	Four months after receiving Division of Drinking Water and comments
Operations and Maintenance Manual	12 months (July 1, 2027)
Climate Change Adaptation Plan	24 months (July 1, 2028)
Salinity Pollution Prevention Plan	24 Months (July 1, 2028)
Groundwater Monitoring Plan	6 months (January 1, 2027)
Surface Water Monitoring Plan	120 days
Time Schedule Compliance Plan	24 months (July 1, 2028)

- [1] Reports are due within the time specified after the effective date of the Notice of Applicability to SCRWA (July 1, 2026).

10.2. Final Title 22 Engineering Report

10.2.1. SCRWA submitted a draft Title 22 engineering report on March 13, 2026, which as of the date of this monitoring and reporting program is under review by the Division of Drinking Water. SCRWA must submit a final Title 22 engineering report in accordance with title 22, article 7, section 60323. Following submittal of the final Title 22 engineering report, Division of Drinking Water will issue a conditional acceptance letter that will specify critical site-specific requirements for monitoring and operation.

10.2.1.1. The engineering report must be modeled after the Division of Drinking Water's June 2023 *Guidelines for the Preparation of an Engineering Report for the Production, Distribution and Use of Recycled Water*.¹⁴

10.2.1.2. The engineering report must be prepared by a qualified engineer licensed in California and experienced in the field of wastewater treatment and shall contain a description of the design of the proposed reclamation system. The report must clearly indicate the means for compliance with the Notice of Applicability, this monitoring and reporting program, and Title 22 Water Recycling Criteria.

10.2.1.3. The engineering report must contain a contingency plan that will ensure that no untreated or inadequately treated wastewater will be delivered to any use areas.

10.3. Operations and Maintenance Manual

10.3.1. In addition to the required components specified in Standard Provisions¹⁵ A.12 and A.28 (and any updates to the Standard Provisions) and General Permit section VI.A.2, the operations and maintenance manual must contain the following components. Each component must contain an implementation schedule and identification of adequate funding to implement the component.

10.3.2. **Sampling and Analysis Plan** – SCRWA's operation and maintenance manual must contain a sampling and analysis plan that meets the requirements specified herein. Anyone performing sampling on behalf of

¹⁴ Guidelines for the Preparation of an Engineering Report for the Distribution and Use of Recycled Water: [Guidelines for the Preparation of an Engineering Report for the Production, Distribution and Use of Recycled Water](#)

¹⁵ See Standard Provisions, 2013: [Standard Provisions 2013](https://www.waterboards.ca.gov/centralcoast/board_decisions/do-cs/wdr_standard_provisions_2013.pdf)
https://www.waterboards.ca.gov/centralcoast/board_decisions/do-cs/wdr_standard_provisions_2013.pdf

SCRWA must be familiar with the sampling and analysis plan. At a minimum, the sampling and analysis plan must contain the following:

- 10.3.2.1. A wastewater treatment process flow schematic with the monitoring locations labeled and scaled Wastewater System maps with treatment components, discharge locations (both treated wastewater and non-potable recycled water), monitoring locations, groundwater wells, storage locations (e.g., chemical, sludge, emergency overflow ponds), and buildings (if SCRWA needs to update Figure 4 to comply with this requirement, a copy of the updated process flow schematic will also be included with the annual report).
- 10.3.2.2. Sample identification details in tabular format. The table must include the sample titles, GeoTracker field point information, sample description(s), and sampling frequencies.
- 10.3.2.3. Sample chain-of-custody procedures and documentation.
- 10.3.2.4. Sample handling/preservation procedures.
- 10.3.2.5. A description of the analytical methods.
- 10.3.2.6. A description of sample containers, preservatives, and holding times.
- 10.3.2.7. For water supply monitoring, a description of the location and method of data collection (e.g., onsite well sampling, use of consumer confidence report).
- 10.3.2.8. For groundwater monitoring, a description of the well purging and field methods.
- 10.3.3. **Wastewater Disposal Management Plan** – SCRWA's operation and maintenance manual must contain a wastewater disposal management plan that is sufficient to ensure compliance with the terms of the General Permit and the Notice of Applicability. At a minimum, the wastewater disposal management plan must include:
 - 10.3.3.1. A description of the wastewater disposal area and a map denoting acreage.
 - 10.3.3.2. Loading calculations based on flow volumes, applied acreage, and biochemical oxygen demand, salts (total dissolved solids, sodium, chloride, sulfate, boron), and nitrogen analytical results.
 - 10.3.3.3. A description of wastewater disposal and water quality protection practices.
- 10.3.4. **Spill Prevention and Emergency Response Plan** – SCRWA's operation and maintenance manual must contain a spill prevention and emergency response plan that is sufficient to ensure compliance with the terms of the General Permit and the Notice of Applicability. The spill prevention and

emergency response plan must describe operation and maintenance activities to prevent accidental releases of wastewater and to effectively respond to such releases and minimize the environmental impact. At a minimum, the spill prevention and emergency response plan must address the following:

- 10.3.4.1. **Operation and Control of Wastewater System** - A description of the wastewater treatment equipment, operational controls, flow measurement and calibration procedures, and treatment system schematic including valve/gate locations.
- 10.3.4.2. **Sludge Handling** - A description of the sludge handling equipment, operational controls, and disposal procedures.
- 10.3.4.3. **Collection System Maintenance** - A description of collection system cleaning and maintenance, equipment tests, and alarm functionality tests to minimize the potential for wastewater spills originating in the collection system or headworks. For collection systems subject to State Water Board Order WQ 2022-0103-DWQ, Statewide Waste Discharge Requirements General Order for Sanitary Sewer Systems (or its replacement), reports prepared to comply with State Water Board Order WQ 2022-0103-DWQ satisfy this requirement.
- 10.3.4.4. **Emergency Response** - A description of emergency response procedures, including for emergencies such as power outage, severe weather, flooding, or inadequate freeboard (for systems with wastewater treatment, storage, or disposal ponds or treated non-potable recycled water storage ponds). An equipment and telephone list for contractors/consultants, emergency personnel, and equipment vendors.
- 10.3.4.5. **Finance** - At a minimum, discuss current fees, projected fees, current budget for spill prevention and emergency response, projected budget for spill prevention and emergency response.
- 10.3.4.6. **Notification Procedures** - Coordination procedures with fire, police, Governor's Office of Emergency Services (CalOES), Central Coast Water Board, and local county health department personnel.
- 10.3.4.7. **Training Records Log** – SCRWA's operation and maintenance manual must contain updated training records logs that demonstrates SCRWA is complying with General Permit section VI.B.3.

10.4. Climate Change Adaptation

- 10.4.1. The Climate Change Adaptation Plan must, at a minimum, include the following components:

10.4.1.1. **Hazards and Vulnerabilities** – Identify climate change hazards, at a minimum accounting for the hazards listed below, applicable to the Wastewater System. Using up-to-date tools, data, and guidance from the State of California (e.g., Cal-Adapt,¹⁶ Sea-Level Rise Guidance from Ocean Protection Council, reports from the Climate-Safe Infrastructure Working Group, the Climate Adaptation Planning Guide, and California Climate Assessment Regional Reports), assess the Wastewater System’s vulnerability to identified hazards that could cause reduction, loss, or failure of treatment processes and/or critical structures at the Wastewater System. Identify and justify the resources (e.g., models and tools, design parameters) used to inform identification of these hazards and vulnerabilities.

10.4.1.1.1. Sea Level Rise – Saltwater intrusion, flooding and inundation, and increased coastal erosion.

10.4.1.1.2. Precipitation Pattern Changes.

10.4.1.1.3. Drought – Decreased influent quantity and quality.

10.4.1.1.4. Peak Events – Flooding and increased influent quantity.

10.4.1.1.5. Temperature fluctuations and extremes.

10.4.1.1.6. Increased wildfires.

10.4.1.1.7. Increased power outages.

10.4.1.2. **Resiliency Actions** – Identify actions to build Wastewater System and operational resilience to identified vulnerabilities, accounting for options that minimize resource impacts.

10.4.1.3. **Adaptation Strategy** – For wastewater systems with design flows over 1 MGD, develop and implement a strategy to complete resiliency actions, at a minimum encompassing the following:

10.4.1.3.1. Prioritization – Prioritized resiliency actions based on risks to water quality but also accounting for costs and benefits.

10.4.1.3.2. Schedule and Milestones – Timeframes to complete prioritized resiliency actions and/or climate change hazard triggers to inform when SCRWA must implement actions. Milestones to complete critical steps for prioritized resiliency actions,

¹⁶ Cal-Adapt is an online resource with downscaled climate project data. It provides users with projections and more detailed downloadable data supporting a range of needs and array of climate models and emissions scenarios. Cal-Adapt offers climate projections for the major stressors facing California, including the following: temperature averages and extremes, precipitation averages and extremes, sea-level rise, wildfires, and drought. The Governor’s Office of Planning and Research (OPR) recommends agencies use Representative Concentration Pathway (RCP) 8.5 for analyses considering impacts through 2050, because there are minimal differences between emissions scenarios during the first half of the 21st century.

designed to demonstrate measurable progress at a steady, or accelerated, completion pace over the established timeframes.

10.4.1.3.3. Financial Planning – Projected costs necessary to implement and sustain resiliency actions and strategy to procure funds.

10.4.1.4. **Recycled Water Feasibility Plan** – Dischargers with Wastewater System design flows over 1 MGD must include a recycled water feasibility plan. The recycled water feasibility plan will assess the viability of using the Wastewater System’s treated wastewater effluent for beneficial reuse including, but not limited to the following:

10.4.1.4.1. Beneficial Reuse Options – Identification of reuse opportunities for SCRWA’s treated effluent that would achieve the highest beneficial impact and best uses possible of non-potable recycled water. Include assessment of the following non-potable recycled water benefits identified in the State Water Board Recycled Water Policy: providing safe alternatives to fresh water or potable water for approved uses; supporting sustainable groundwater and surface water uses with the intent of substituting use of treated effluent for use of fresh water or potable water; and diversifying community water supplies and mitigating for the impacts of climate change.¹⁷

10.4.1.4.2. Viable Users – Identification and viability evaluation of all potential users of the Wastewater System’s treated effluent associated with identified beneficial reuse options. SCRWA must demonstrate it engaged potential water purveyors and customers about the viability of reusing the Wastewater System’s treated effluent.

10.4.1.4.3. Infrastructure Upgrades – Assessment of infrastructure needs to produce non-potable recycled water for identified beneficial reuses and convey non-potable recycled water to beneficial reuse locations.

10.4.1.4.4. Fiscal Analysis – Cost estimates for identified reuse options, at a minimum including project development, construction, and long-term maintenance and lifecycle costs. This analysis shall also include identification and evaluation of the following: 1) funding options, including at a minimum, revenues from recycled water sales, grant and loan funding opportunities, and financing from other benefiting parties and 2) funding limitations.

10.4.1.4.5. Schedule and Milestones for Next Steps – Timeframes to assess the feasibility of identified beneficial reuse options and

¹⁷ [Water Quality Control Policy for Recycled Water](#), State Water Board, adopted December 11, 2018, page 1.

to identify proposed beneficial reuse options based on prioritized water recycling and reuse opportunities, feasibility assessments, and other factors identified by SCRWA.

10.4.1.4.6. Identification of implementation schedule and adequate funding to implement the plan.

10.5. Salinity Pollution Prevention Plan

10.5.1. SCRWA must submit a salinity pollution prevention plan to ensure that the overall impact of treated wastewater disposal and non-potable water recycling does not degrade groundwater quality. At a minimum, the salinity pollution prevention plan must address the following:

10.5.1.1. A description of implementable salt management actions that reduce mass loading of salts in treated effluent to protect beneficial uses of groundwater.

10.5.1.1.1. A description of salt reduction measures that identify and address all potential salt contributors to the collection system (e.g., water supply, commercial, industrial, residential) and wastewater treatment process.

10.5.2. SCRWA must participate in regional salt and nutrient management planning and implement appropriate measures identified in resulting plans and programs.

10.6. Time Schedule Compliance Plan

10.6.1. As set forth in General Permit sections V.A and VI.A.5, if SCRWA anticipates that additional time is needed to achieve compliance with the limitations, SCRWA must prepare and submit for Central Coast Water Board review and approval a time schedule compliance plan. At a minimum, the time schedule compliance plan must address the following:

10.6.2. Comparison of the current effluent quality to the effluent and groundwater limitations in General Permit Table 3-6.

10.6.3. A detailed description and chronology of efforts, since issuance of the Notice of Applicability, to reduce wastes.

10.6.4. Justification of the need for additional time to achieve the effluent or groundwater limitations in General Permit Table 3-6.

10.6.5. A detailed time schedule of specific actions SCRWA will take to achieve the effluent or groundwater limitations.

10.6.6. A demonstration that the time schedule requested is as short as possible, considering the technological, operation, and economic factors that affect

the design, development, and implementation of the measures that are necessary to comply with the effluent or groundwater limitations.

10.6.7. If the requested time schedule exceeds one year, the proposed schedule shall include interim requirements and the dates for their achievement. The interim requirements shall include both of the following:

10.6.7.1. Effluent or groundwater limitations for the pollutants of concern.

10.6.7.2. Actions, measurable milestones, and tangible products leading to compliance with the effluent or groundwater limitations.

11. SCRWA Legal Requirements

11.1. The Central Coast Water Board requires SCRWA to submit the technical and monitoring reports described in this monitoring and reporting program pursuant to [California Water Code section 13267](#). Failure to submit reports in accordance with the schedules established by the Permit or failure to submit a report of sufficient technical quality to be acceptable to the Central Coast Water Board may subject SCRWA to enforcement action pursuant to [California Water Code section 13268](#).

11.2. The burden, including costs, of the reports bears a reasonable relationship to their need and the benefits to be obtained. The requirement for the reports is necessary to ensure compliance with the Permit and to protect water quality. SCRWA is required to submit this information because it is subject to the Permit and is responsible for the discharge.

11.3. SCRWA must implement the above monitoring program on July 1, 2026. Until July 1, 2026, SCRWA must continue to implement monitoring and reporting required by NPDES permit Order R3-2017-0028 and Master Water Reclamation Requirements Order 98-052. The Central Coast Water Board may rescind or modify the monitoring and reporting program at any time. SCRWA must not implement any changes to this monitoring and reporting program unless and until a revised monitoring and reporting program is issued by the Central Coast Water Board.

Ordered By:

for Ryan E. Lodge

Executive Officer

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rev: 11/13/24

ATTACHMENT 4
DIVISION OF DRINKING WATER CORRESPONDENCE RELATED TO UV
DISINFECTION

State Water Resources Control Board

Division of Drinking Water

December 23, 2015

Mr. Saeid Vaziry
South County Regional Wastewater Authority
1500 Southside Drive
Gilroy, CA 95020

XYLEM/WEDECO TAK-55 UV DISINFECTION ENGINEERING REPORT SOUTH COUNTY REGIONAL WASTEWATER AUTHORITY, SYSTEM NO. 4390002

Dear Mr. Vaziry:

This letter is in response to the revised *Engineering Report for the Production, Distribution and Use of Recycled Water: Ultraviolet Disinfection Facility*, dated October 28, 2015. South County Regional Wastewater Authority (SCRWA) submitted the response in order to address three main areas of concern following submission of the first engineering report: plant reliability features, operations and maintenance manual including reliability features, and check point bioassay test protocol. The Division cannot complete the review and recommended applicable performance criteria to the Regional Board to incorporate into the final permit for the UV system based on the information provided thus far. The Division has the following comments.

Operations and Maintenance Manual

The plant must be operated in accordance with an approved operations plan, which specifies clearly the operational limits and responses required for critical alarms. The operations plan must be submitted and approved prior to issuance of the operating permit. A copy of the approved operations plan should be maintained at the treatment plant and be readily available to operations personnel and regulatory agencies. The operations plan should be easy to read and operator-friendly. A quick reference plant operations data sheet should be posted at the treatment plant and include the following information:

- a. The alarm set points for flow and UVT.
- b. The values of flow, UV dose, and UVT when effluent must be diverted to waste.
- c. The required frequency of verification and calibration for all meters/analyzers measuring flow and UV transmittance.
- d. The required frequency of mechanical cleaning and equipment inspection.
- e. The UV lamp tracking procedures and replacement intervals.

SCRWA submitted copies of manufacture equipment specifications and has not met the intent of an Operations and Maintenance Manual. SCRWA should submit an Operations and Maintenance Manual that is sufficiently detailed that an experienced operator could operate the facility with minimal training.

Commissioning of the UV System

On-site commissioning must be done as outlined in the 2012 NWRI UV guidelines. The commissioning must be performed by an independent third-party, experienced in UV disinfection of recycled water and qualified to perform the study. "Spot check" tests must be conducted by a PE in CA qualified and experienced in this testing. The protocol should be revised and re-submitted with the name of this person.

The proposed spot-check testing only goes down to 65% UVT. There should be a test at 55% UVT at 4.5 MGD. The protocol should be revised and re-submitted including this revision.

After completion of testing, whether good or bad, the results must be submitted for review.

The commissioning protocol should determine that the flow between the UV channels is split evenly. If not, the flow has to be controlled within bounds or the results may be skewed. Flow split measurements should be made before the bioassay. Control of the flow and water level should be carefully considered. If the flow is not even, adjustments will need to be made to the UV dose control equation to compensate.

The protocol should propose a maximum water level appropriate for the operational UV dose equation programed into SCADA. The commissioning protocol should clarify whether the UV system can control the level to not exceed the maximum water level. The commissioning protocol should include a check of the water level for each channel.

A fouling study should be conducted to determine the minimum frequency of sleeve cleaning/wiping.

The protocol should include continuous monitoring and recording of the flow rate, UVT, electrical supply voltage, power input to the UV lamps, and UV sensor signals.

At each flow rate, at a minimum, water temperature, water level, and headloss across each UV reactor should be measured.

UV sensors should monitoring UV intensity continuously and be sensitive enough to detect changes in UVT, ballast output, sleeve fouling, and lamp aging.

The protocol should define where the MS-2 samples will be taken from.

After completion of testing, the final report should include calibration certification of the UV sensor, flow meter, and power meter.

Monitoring and Reporting

As stated previously, the Engineering Report refers to the 2010 NPDES permit covering river discharge and recycled water use to describe monitoring and reporting requirements. The Division finds that the monitoring and reporting requirements listed in the 2010 NPDES permit for existing

facilities do not adequately address monitoring and reporting procedures that will need to be implemented at the new UV disinfection treatment facility. Records should be maintained on lamp age and lamp replacement and at a minimum, the following monitoring should be completed and submitted monthly to the Board:

1. Results of daily coliform bacteria monitoring, running 7-day median calculation, maximum daily coliform reading for previous month(s), and minimum daily chlorine residual.
2. Daily maximum turbidity reading.
3. Daily UV compliance determinations, including minimum daily UV dose.
4. Summary of operational problems, plant and equipment breakdowns and diversions to emergency storage or disposal, and all corrective or preventative actions taken.

Treatment Reliability Features and Alarms

The facility is not staffed full-time; however, in the event of an after-hours alarm or shutdown notification an auto-dialer will contact staff on an emergency notification list and a separate auto-dialer will contact the secure communications staffed 24-hours by the City of Gilroy Police Department who will contact the operator on call or the plant manager. The facility is provided with back-up power via on-site diesel engine generators. Pursuant to Section 60337, Title 22, the power supply must be provided with one of the following reliability features:

1. Alarm and standby power source.
2. Alarm and automatically actuated short-term retention or disposal provisions as specified in Section 60341.
3. Automatically actuated long-term storage or disposal provisions as specified in Section 60341

SCRWA should verify which of the aforementioned options is their chosen treatment and reliability feature and how compliance will be met.

If you have any questions regarding this letter, please feel free to contact Ms. Karen Nishimoto at (510) 620-3461.

Sincerely,



Eric Lacy, P.E.
District Engineer, Santa Clara District
Division of Drinking Water
State Water Resources Control Board

cc: Santa Clara County Environmental Health Department

Mr. Blair Allen, P.E.
CRWQCB – San Francisco Region
1515 Clay Street, Suite 1400
Oakland, CA 94612

State Water Resources Control Board
Division of Drinking Water

October 10, 2016

Ms. Katie DiSimone
CRWQCB – Central Coast Region
895 Aerovista Place, Suite 101
San Luis Obispo, CA 93401

SOUTH COUNTY REGIONAL WASTEWATER AUTHORITY – UV DISINFECTION
SOUTH COUNTY REGIONAL WASTEWATER AUTHORITY, SYSTEM NO. 4390002

Dear Ms. DiSimone:

The State Water Resources Control Board Division of Drinking Water (Division) has reviewed the South County Regional Wastewater Authority (SCRWA) UV Spot Check Bioassay Results of the WEDECO TAK 55 UV system dated September 8, 2016. As part of the tertiary treatment system a 360W lamp WEDECO TAK55 Wastewater UV Reactor was installed. As a whole the UV system consists of two channels. Each channel has two duty and one redundant bank. Each bank has four modules and each module has eighteen lamps and an integral sensor. This WEDECO TAK55 utilizes low-pressure high-output (LPHO) 360-W lamps (ELR30).

Spot Check Bioassay

To verify performance of the UV system at several flows and UVTs, an on-site spot check bioassay using seeded MS2, was conducted. Results, documenting virus disinfection performance of the UV system compared to the standards found in Title 22 of the California Code of Regulations, was submitted to the Division for acceptance. The report "Bioassay Testing Results for the SCRWA WEDECO TAK55 UV System Final Draft" (Carollo Engineers, August 2016) contains the bioassay results of the testing on-site at the plant.

Eleven tests were conducted at various flow rates (2.07 to 4.65 MGD per channel) and UVTs (60% to 72%). These eleven "spot checks" were compared to the dose predicted by the operating equations developed and documented in the "Wedeco Open Channel TAK55 Wastewater UV Reactor 360W Validation Report Final, Carollo Engineers dated January 2010". For only four of the eleven test runs (12.5%), the dose measured was greater than or equal to the dose predicted by the dose operating equation, which controls the power and dosage level applied. The Spot Check Bioassay Test Report indicates that the UV system tested did not perform as expected and should be operated to deliver a minimum UV dose of 110 mJ/cm² at all times.

Water Level Measurements

The report states, "WEDECO recommends that the maximum water depth over the top of the UV lamps should not exceed approximately 1.0 to 1.2 inches. Table 4 shows the results of water level measurements taken during bioassay testing. In some cases, the maximum water level is greater than recommended." The water level testing indicates that most of the issues are at flow rates exceeding 4 MGD in a channel, including a high of 1.7 inches. This leads to inadequate disinfection; therefore, it is recommended that the flow should be limited to 4 MGD.

The Division recommends the following permit provisions for the UV system. The following recommendations are based on the equipment cited in the report.

1. Each UV channel must be operated independently to deliver a minimum UV dose of 110 mJ/cm² at all times.
2. The equations below must be used for each UV reactor as part of the automatic UV disinfection control system for calculating UV dose and should be specified as a permit provision. They are from the "Wedeco Open Channel TAK55 Wastewater UV Reactor 360W Validation Report Final, Carollo Engineers, dated January 2010".

$$S_o = e^{-0.63} \times e^{0.0452 \times \text{UVT}} \times \text{Power}^{0.9124}$$

$$\text{RED}_{\text{calc}} = \text{CR} \times 10^{1.863} \times \text{UVA}^{-1.695} \times [\text{S}/\text{S}_o]^{0.927} \times \text{Q}^{-0.839}$$

Where:

UVT = UV transmittance at 254 nm, expressed as a whole number, e.g., 60 to 72,

S = Measured UV sensor value (mW/cm²)

S_o = Calculated intensity from new lamp at full power (at same UVT) with clean sleeves, typically expressed as a function of UVT (mW/cm²).

RED_{calc} = UV dose calculated per bank (mJ/cm²)

CR = Confidence factor = 0.938

UVA = UV absorbance at 254 nm (cm⁻¹), e.g., between 0.222 and 0.142¹.

Q = Flow rate, calculated as gpm divided by the number of lamps in one bank (gpm/lamp)

Power = Measured ballast power setting (kW)

3. The SCRWA UV disinfection system is limited to the following operational parameter ranges:
 - a. Permit total plant flow up to 8 MGD (4.0 MGD per UV channel).
 - b. UVTs at or above 60 percent,
 - c. UV sensor intensities ranging from 2.0 to 4.3 mW/cm².
4. On-line monitoring of UV intensity, flow, and UVT must be provided at all times.

¹ At UVT values above 72 percent, the value (72 UVT, or UVA = 0.142) should be used as the default value in the S_o and RED calculations respectively.

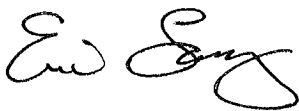
5. Flow meters, UV intensity sensors, and UVT monitors must be properly calibrated to ensure proper disinfection.
6. At least monthly, all duty UV intensity sensors must be checked for calibration against a reference UV intensity sensor.
7. For all UV intensity sensors in use, the ratio of the duty UV sensor intensity to the reference UV sensor intensity must be less than or equal to 1.2. If the calibration ratio is >1.2 , the failed duty UV sensor must be replaced by a properly calibrated sensor and recalibrated by a qualified facility. The reference UV intensity sensors shall be recalibrated at least annually by a qualified facility using a National Institute of Standards and Technology (NIST) traceable standard.
8. UVT meter must be inspected and checked against a reference bench-top unit weekly to document accuracy.
9. If the on-line analyzer UVT reading varies from the bench-top spectrophotometer UVT reading by 2% or more, the on-line UVT analyzer must be recalibrated by a procedure recommended by the manufacturer.
10. Flow meters measuring the flow through a UV reactor must be verified to determine accuracy at least monthly via checking the flow reading against other flow determination methods.
11. Each UV reactor at the UV system must be designed with built-in automatic reliability features that must be triggered by critical alarm setpoints.
12. Conditions triggering an alarm and startup the redundant bank include the following:
 - a. the UV dose goes below 115 mJ/cm^2 , and
 - b. whole bank failure.
13. Conditions that should divert effluent to waste include the following:
 - a. UV dose is below the minimum UV dose of 110 mJ/cm^2 ,
 - b. UVT is below the minimum UVT commissioned of 60%,
 - c. UV intensity below the minimum validated of 2.0 mW/cm^2 ,
 - d. complete UV channel failure, and
 - e. flow above the maximum flow commissioned of 4.0 MGD per channel.
14. The UV system should be operated in accordance with an approved operations plan, which specifies clearly the operational limits and responses required for critical alarms. The operations plan should be submitted and approved prior to issuance of the operating permit. A copy of the approved operations plan should be maintained at the treatment plant and be readily available to operations personnel and regulatory agencies. A quick

reference plant operations data sheet should be posted at the treatment plant and include the following information:

- a. The alarm set points for flow, UV dose, UV intensity, and UVT.
 - b. The values of flow, UV dose, UV intensity, and UVT when effluent must be diverted to waste.
 - c. The required frequency of verification and calibration for all meters/analyzers measuring flow, UV intensity, and UV transmittance.
 - d. The required frequency of mechanical cleaning and equipment inspection.
 - e. The UV lamp hour tracking procedures and replacement intervals.
15. This UV dose equation assumes that the intensity sensors would measure the decline as the lamps age. Since there is one UV Intensity sensor that monitors two of the 72 lamps in a bank, the two lamps with the highest number of hours should be the ones closest to the UV intensity sensor.
16. Equivalent or substitutions of equipment are not acceptable without an adequate demonstration of equivalent disinfection performance.
17. These applicable recommendations should be incorporated into the final permit for the UV system. Approval for the use of any and all water recycling applications is granted through the Regional Water Quality Control Board's Water Reclamation permitting process.

If you have any questions concerning this letter, please contact Karen Nishimoto at (510) 620-3461.

Sincerely,

A handwritten signature in black ink, appearing to read 'Eric Lacy', with a stylized flourish at the end.

Eric Lacy, P.E.
District Engineer, Santa Clara District
Division of Drinking Water
State Water Resources Control Board

cc: Santa Clara County Environmental Health Department

Mr. Saeid Vaziry
South County Regional Wastewater Authority
1500 Southside Drive
Gilroy, CA 95020

State Water Resources Control Board

Division of Drinking Water

February 1, 2018

Ms. Katie DiSimone
CRWQCB – Central Coast Region
895 Aerovista Place, Suite 101
San Luis Obispo, CA 93401

SCRWA WEDECO TAK55 UV 360W SPOT CHECK BIOASSAY SOUTH COUNTY REGIONAL WASTEWATER AUTHORITY, SYSTEM NO. 4390002

Dear Ms. DiSimone:

The State Water Resources Control Board Division of Drinking Water (Division) has reviewed the South County Regional Wastewater Authority (SCRWA) UV Spot Check Bioassay results of the WEDECO TAK 55 UV system submitted July 17, 2017. As part of the tertiary treatment system a 360W lamp WEDECO TAK55 Wastewater UV Reactor was installed. As a whole the UV system consists of two channels. Each channel has two duty and one redundant bank. Each bank has four modules and each module has eighteen lamps and an integral sensor. This WEDECO TAK55 utilizes low-pressure high-output (LPHO) 360-W lamps (ELR30).

Spot Check Bioassay

To verify performance of the UV system at several flows and UVTs, an on-site spot check bioassay using seeded MS2, was conducted. Results, documenting virus disinfection performance of the UV system compared to the standards found in Title 22 of the California Code of Regulations, was submitted to the Division for acceptance. The report "Bioassay Testing Results for the SCRWA WEDECO TAK55 UV System Final Draft" (Carollo Engineers, July 2017) contains the bioassay results of the testing on-site at the plant.

Eight tests were conducted at various flow rates (2.02 to 4.41 MGD per channel) and UVTs (57% to 73%). These eight "spot checks" were compared to the dose predicted by the operating equations developed and documented in the "Wedeco Open Channel TAK55 Wastewater UV Reactor 360W Validation Report Final, Carollo Engineers dated January 2010". For only two of the eight test runs (25%), the dose measured was greater than or equal to the dose predicted by the dose operating equation, which controls the power and dosage level applied. Seven out of the eight test runs should have demonstrated a performance equal to or exceeding the design intent. Therefore, the Spot Check Bioassay Test Report indicates that the UV system tested did not

FELICIA MARCUS, CHAIR | EILEEN SOBECK, EXECUTIVE DIRECTOR

850 Marina Bay Parkway, Bldg. P, 2nd Floor, Richmond, CA 94804-6403 | www.waterboards.ca.gov

perform as expected and should be operated to deliver a minimum UV dose of 112 mJ/cm² at all times.

The Division recommends the following permit provisions for the UV system. The following recommendations are based on the equipment cited in the report.

1. Each UV channel must be operated independently to deliver a minimum UV dose of 112 mJ/cm² at all times.
2. The equations below must be used for each UV reactor as part of the automatic UV disinfection control system for calculating UV dose and should be specified as a permit provision. They are from the "Wedeco Open Channel TAK55 Wastewater UV Reactor 360W Validation Report Final, Carollo Engineers, dated January 2010".

$$S_o = e^{-0.63} \times e^{0.0452 \times \text{UVT}} \times \text{Power}^{0.9124}$$

$$\text{RED}_{\text{calc}} = \text{CR} \times 10^{1.863} \times \text{UVA}^{-1.695} \times [\text{S}/\text{S}_o]^{0.927} \times \text{Q}^{-0.839}$$

Where:

UVT = UV transmittance at 254 nm, expressed as a whole number, e.g., 60 to 72,

S = Measured UV sensor value (mW/cm²)

S_o = Calculated intensity from new lamp at full power (at same UVT) with clean sleeves, typically expressed as a function of UVT (mW/cm²).

RED_{calc} = UV dose calculated per bank (mJ/cm²)

CR = Confidence factor = 0.938

UVA = UV absorbance at 254 nm (cm⁻¹), e.g., between 0.222 and 0.142¹.

Q = Flow rate, calculated as gpm divided by the number of lamps in one bank (gpm/lamp)

Power = Measured ballast power setting (kW)

3. The SCRWA UV disinfection system is limited to the following operational parameter ranges:
 - a. Permit total plant flow up to 9 MGD (4.5 MGD per UV channel).
 - b. UVTs at or above 60 percent,
 - c. UV sensor intensities ranging from 2.0 to 4.3 mW/cm².
4. On-line monitoring of UV intensity, flow, and UVT must be provided at all times.
5. Flow meters, UV intensity sensors, and UVT monitors must be properly calibrated to ensure proper disinfection.

¹ At UVT values above 72 percent, the value (72 UVT, or UVA=0.142) should be used as the default value in the S_o and RED calculations respectively.

6. At least monthly, all duty UV intensity sensors must be checked for calibration against a reference UV intensity sensor.
7. For all UV intensity sensors in use, the ratio of the duty UV sensor intensity to the reference UV sensor intensity must be less than or equal to 1.2. If the calibration ratio is >1.2 , the failed duty UV sensor must be replaced by a properly calibrated sensor and recalibrated by a qualified facility. The reference UV intensity sensors shall be recalibrated at least annually by a qualified facility using a National Institute of Standards and Technology (NIST) traceable standard.
8. UVT meter must be inspected and checked against a reference bench-top unit weekly to document accuracy.
9. If the on-line analyzer UVT reading varies from the bench-top spectrophotometer UVT reading by 2% or more, the on-line UVT analyzer must be recalibrated by a procedure recommended by the manufacturer.
10. Flow meters measuring the flow through a UV reactor must be verified to determine accuracy at least monthly via checking the flow reading against other flow determination methods.
11. Each UV reactor at the UV system must be designed with built-in automatic reliability features that must be triggered by critical alarm setpoints.
12. Conditions triggering an alarm and startup the redundant bank include the following:
 - a. the UV dose goes below 118 mJ/cm^2 ,
 - b. whole bank failure,
13. Conditions that should divert effluent to waste include the following:
 - a. UV dose is below the minimum UV dose of 112 mJ/cm^2 ,
 - b. UVT is below 55%,
 - c. UV intensity below the minimum validated of 2.0 mW/cm^2
 - d. complete UV channel failure, and
 - e. flow above 4.5 MGD per channel.
14. The UV system should be operated in accordance with an approved operations plan, which specifies clearly the operational limits and responses required for critical alarms. The operations plan should be submitted and approved prior to issuance of the operating permit. A copy of the approved operations plan should be maintained at the treatment plant and be readily available to operations personnel and regulatory agencies. A quick reference plant operations data sheet should be posted at the treatment plant and include the following information:
 - a. The alarm set points for flow, UV dose, UV intensity, and UVT.

- b. The values of flow, UV dose, UV intensity, and UVT when effluent must be diverted to waste.
 - c. The required frequency of verification and calibration for all meters/analyzers measuring flow, UV intensity, and UV transmittance.
 - d. The required frequency of mechanical cleaning and equipment inspection.
 - e. The UV lamp hour tracking procedures and replacement intervals.
15. This UV dose equation assumes that the intensity sensors would measure the decline as the lamps age. Since there is one UV Intensity sensor that monitors two of the 72 lamps in a bank, the two lamps with the highest number of hours should be the ones closest to the UV intensity sensor.
16. Equivalent or substitutions of equipment are not acceptable without an adequate demonstration of equivalent disinfection performance.
17. These applicable recommendations should be incorporated into the final permit for the UV system. Approval for the use of any and all water recycling applications is granted through the Regional Water Quality Control Board's Water Reclamation permitting process.

If you have any questions regarding this letter, please feel free to contact Ms. Karen Nishimoto at (510) 620-3461.

Sincerely,



Eric Lacy, P.E.
District Engineer, Santa Clara District
Division of Drinking Water
State Water Resources Control Board

cc: Santa Clara County Environmental Health Department

Mr. Saeid Vaziry
South County Regional Wastewater Authority
1500 Southside Drive
Gilroy, CA 95020

State Water Resources Control Board

February 6, 2019

Ms. Sheila Soderberg
CRWQCB – Central Coast Region
895 Aerovista Place, Suite 101
San Luis Obispo, CA 93401

Dear Ms. Soderberg:

SCRWA UV DISINFECTION FACILITY South County Regional Wastewater Authority, System No. 4390002

This letter is in regards to the South County Regional Wastewater Authority (SCRWA) ultra violet (UV) disinfection facility and the recycling specifications listed within Order No. R3-2017-0028. As part of the tertiary treatment system, SCRWA installed a Xylem/WEDECO TAK-55 HP UV disinfection system that consists of two in-channel disinfection trains. Each channel contains three banks of UV lamps in series (2 duty, 1 standby). Each bank houses 4 modules each with 18 horizontal UV lamps. The UV system was designed to treat up to 4.5 million gallons per day (MGD) per channel, for a total of 9 MGD. To verify the performance of the UV system at several flow rates and UVTs, an on-site spot check bioassay using seeded MS2 was conducted in April 2016. Results of the bioassay was submitted to the Division of Drinking Water (Division) for review. Upon review, the Division issued a letter to the California Regional Water Quality Control Board (CRWCB) – Central Coast Region on October 10, 2016 with recommended permit provisions for the UV system. In an effort to understand the underperformance of the UV system, WEDECO reviewed the bioassay data, conducted an onsite investigation of the potential causes and identified potential strategies to mitigate the performance issues. A follow-up spot check bioassay was conducted in March 2017 by WEDECO and Carollo Engineers. The "Bioassay Testing Results for the SCRWA WEDECO TAK55 UV System" report dated July 2017 was submitted to the Division for review. Based on the second spot-check bioassay results, the Division issued a letter to the CRWCB on February 1, 2018 with revised recommended permit provisions.

To incorporate the revised operating conditions for the SCRWA UV facility, the Division recommends the following revisions to the recycling specifications listed within Order No. R3-2017-0028. The revisions are noted below in red font.

C. Recycling Specifications – Discharge Point No. 003

12. Recycled water disinfected with ultraviolet (UV) light shall comply with the following DDW recommendations:

- a. Each UV channel must be operated independently to deliver a minimum UV dose of ~~110 mJ/cm²~~ **112 mJ/cm²** at all times.
- b. The equations below must be used for each UV reactor as part of the automatic UV disinfection control system for calculating UV dose. They are from the ~~"Wedeco Open Channel TAK55 Wastewater UV Reactor 360W Validation Report Final," Carollo Engineers, dated July 2010~~ **"Bioassay Testing Results for the SCRWA WEDECO TAK55 UV System", final report dated July 2017.**

$$RED_{calc} = CR \times 10^{1.863} \times UVA^{-1.695 - 1.615} \times [S/S_0]^{0.927} \times Q^{-0.839}$$

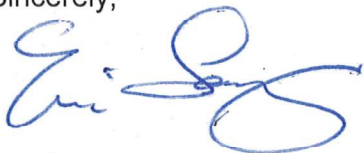
- c. The UV disinfection system is limited to the following operational parameter ranges:
 - i. Total plant flow is limited to ~~8~~ **9** MGD (~~4.0~~ **4.5** MGD per UV channel). ~~Flow may be increased to 9 MGD total in accordance with DDW recommendations and Executive Officer approval.~~
 - l. Conditions triggering an alarm and startup of the redundant bank include the following:
 - i. The UV dose goes below ~~115 mJ/cm²~~ **118 mJ/cm²**, and
 - m. Conditions that require diverting effluent to waste include the following:
 - i. UV dose is below the minimum UV dose of ~~110 mJ/cm²~~ **112 mJ/cm²**.
 - ii. UVT is below the minimum UVT commissioned of ~~60%~~ **55%**.
 - v. Flow above the maximum flow commissioned of ~~4.0~~ **4.5** MGD per channel.

The Division has also reviewed the SCRWA Ultraviolet Disinfection Facility Operations Plan dated January 2019 and find it acceptable. We recommend modifying recycling specifications 12n to the following:

- n. The UV system must be operated in accordance with the approved operations plan, dated January 2019, that specifies clearly the operational limits and responses required for critical alarms. Any updates or modifications to the operations plan must be submitted to and approved by DDW prior to implementation.

We would appreciate a copy of the revised Order upon issuance. If you have any questions regarding this letter, please contact Ms. Van Tsang at (510) 620-3602.

Sincerely,



Eric Lacy, P.E.
District Engineer
Santa Clara District
Division of Drinking Water
State Water Resources Control Board

cc: Santa Clara County Environmental Health

Mr. Saeid Vaziry – SCRWA (via email)

Mr. Billy Wong – Stantec (via email)

State Water Resources Control Board

Division of Drinking Water

December 8, 2020

Ms. Jennifer Epp
Senior Water Resource Control Engineer
CRWQCB – Central Coast Region
895 Aerovista Place, Suite 101
San Luis Obispo, CA 93401

Dear Ms. Epp:

SOUTH COUNTY REGIONAL WASTEWATER AUTHORITY – UV DISINFECTION

South County Regional Wastewater Authority, System No. 4390002

This letter is in regards to the South County Regional Wastewater Authority (SCRWA) ultra violet (UV) disinfection facility and the recycling specifications listed within Order No. R3-2017-0028 (Order). As part of the tertiary treatment system, SCRWA installed a Xylem/WEDECO TAK-55 HP UV disinfection system that consists of two in-channel disinfection trains. Each channel contains three banks of UV lamps in series (2 duty, 1 standby). Each bank houses 4 modules each with 18 horizontal UV lamps. The UV system was designed to treat up to 4.5 million gallons per day (MGD) per channel, for a total of 9 MGD. The State Water Resources Control Board – Division of Drinking Water (Division) issued a letter to the California Regional Water Quality Control Board (CRWQCB) – Central Coast on October 10, 2016 with recommended provisions for the UV system based on results from the on-site spot check bioassay conducted in April 2016. In an effort to understand the underperformance of the UV system, a follow-up spot check bioassay was conducted in March 2017. As a result, revised operating conditions were recommended to CRWQCB on February 6, 2019. Since then, the Division has further reviewed the Order and results from the follow-up spot check bioassay and offer the following modification to provision 12.c.iii on page 8 of Order to accurately reflect the testing conditions and UV intensity values recorded during the March 2017 spot check bioassay. The recommended revision is noted below in red font.

12. Recycled water disinfected with ultraviolet light shall comply with the following Division recommendations:
 - c. The UV disinfection system is limited to the following operational parameter ranges:
 - iii. UV sensor intensities ranging from 2.0 to 4.3 **5.3** mW/cm².

E. JOAQUIN ESQUIVEL, CHAIR | EILEEN SOBECK, EXECUTIVE DIRECTOR

UV intensity measured above the upper limit should be considered as 5.3 mW/cm^2 when used to calculate the UV dose as intensity values greater than 5.3 mW/cm^2 were not validated for accuracy during the March 2017 spot check bioassay.

The Division would appreciate a copy of the revised Order once issued to incorporate the revision noted above and within the February 6, 2019 letter. If you have any questions regarding this letter, please contact Ms. Van Tsang at (510) 620-3602.

Sincerely,

A handwritten signature in blue ink, appearing to read "Eric Lacy", with a stylized flourish at the end.

Eric Lacy, P.E.
District Engineer
Santa Clara District
Division of Drinking Water
State Water Resources Control Board

Cc: Santa Clara County Environmental Health Department

[EXTERNAL] SCRWA UV Disinfection Demonstration videos

Bernados, Brian@Waterboards <Brian.Bernados@waterboards.ca.gov>

Thu 6/24/2021 11:26 AM

To: Packard, Harvey@Waterboards <Harvey.Packard@waterboards.ca.gov>; DiSimone, Katie@Waterboards <Katie.DiSimone@waterboards.ca.gov>
Cc: Barnard, Randy@Waterboards <Randy.Barnard@waterboards.ca.gov>; Amah, Ginachi@Waterboards <Ginachi.Amah@waterboards.ca.gov>; Ortiz, Aide@Waterboards <Aide.Ortiz@waterboards.ca.gov>; Lacy, Eric@Waterboards <Eric.Lacy@waterboards.ca.gov>; Tsang, Van@Waterboards <Van.Tsang@waterboards.ca.gov>; Vasquez, Christopher/GIL <Christopher.Vasquez1@jacobs.com>; Saeid Vaziry <saeid.vaziry@ci.gilroy.ca.us>; Edgar, John/GIL <John.Edgar@jacobs.com>

This is formal notice that the Division of Drinking Water (DDW) finds that the South County Regional Wastewater Authority (SCRWA) UV disinfection system is acceptable for use as "disinfected tertiary recycled water" in accordance with Title 22, Section 60301.230.

On June 23, 2021, DDW staff witnessed a virtual presentation of critical alarms testing at the SCRWA UV Disinfection Facility located in Gilroy, CA. As part of the tertiary treatment system a 360W lamp WEDECO TAK55 Wastewater UV Reactor was installed. As a whole the UV system consists of two channels. Each channel has two duty and one redundant bank. Each bank has four modules and each module has eighteen lamps and an integral sensor. This WEDECO TAK55 utilizes low-pressure high-output (LPHO) 360-W lamps (ELR30).

DDW staff requested demonstration of the critical alarm conditions, which divert plant effluent to waste:

1. Alarm and startup the redundant bank if the UV dose goes below 118 mJ/cm²,
2. Alarm to divert effluent to waste if UV dose is below the minimum UV dose of 112 mJ/cm²
3. Alarm to divert effluent to waste if UV intensity below the minimum validated of 2.0 mW/cm²
4. Alarm divert effluent to waste if flow above 4.5 MGD per channel.

SCRWA successfully demonstrated that the critical wastewater treatment plant alarms are functional for its operations. Upon critical alarms being triggered, recycled water in the effluent channels is automatically routed to the chlorine contact basins for disinfection. SCRWA stated that the alarm setpoints are controlled by the UV manufacturer WEDECO and can't be changed by an operator.

If you have any questions, please contact me.
Brian Bernados, DDW Technical Specialist